DELHI SUBORDINATE SERVICES SELECTION BOARD (DSSSB) EXAMINATION GUIDE

Research-Based DSSSB Arithmetic Aptitude Cheat Sheet

Complete Section-A Arithmetical & Numerical Ability Reference, Verified Previous-Year Question Analysis, 21 Topic Modules, Shortcut Tables & Full Mock Test

1. Examination Profile & Target Post Specification

Examination Parameter	Verified Official Specification & Scope Disclosure
Conducting Authority	Delhi Subordinate Services Selection Board (DSSSB), Govt. of NCT of Delhi (https://dsssb.delhi.gov.in/)
Target Examination & Posts	**Primary Target:** DSSSB Trained Graduate Teacher (TGT) — One-Tier (Technical/Teaching) CBT across all subjects (Maths, Natural Science, Social Science, English, Hindi, Sanskrit, Computer Science, Special Education Teacher). **Comparable General One-Tier Posts Disclosed:** Primary Teacher (PRT), Assistant Teacher, Lower Division Clerk (LDC), Junior Assistant, Multi-Tasking Staff (MTS), Patwari, and Warder, which share the exact same official Section-A Arithmetical & Numerical Ability syllabus.
Section-A Arithmetic Weightage	**DSSSB TGT / PRT / PGT Posts:** 20 Questions = 20 Marks (out of 200 total marks; Section A = 100 marks across 5 sections × 20 Qs; Section B = 100 marks Concerned Subject & Pedagogy). **DSSSB LDC / Jr. Assistant / MTS (General One-Tier):** 35 to 40 Questions = 35 to 40 Marks (out of 200 marks).
Marking Scheme & Duration	**Correct Answer:** +1.00 Mark | **Negative Marking:** −0.25 Mark (1/4th penalty per wrong MCQ) | **Unattempted:** 0 Marks. **Total Composite Duration:** 2 Hours (120 Minutes) for 200 Questions. Recommended time allocation for TGT Section-A Arithmetic (20 Qs): **18 to 22 minutes** (~55–65 seconds per question).
Official Syllabus Level & Language	**Difficulty Level:** Secondary (Class 10th) foundation scaling from direct formula application to multi-step competitive CBT word problems. **Explanation Style:** Simple, step-by-step Class-10 English with both full algebraic derivations and verified exam shortcuts.

2. Table of Contents & Document Architecture

Section	Module Title & Contents	Scope / Priority
Part I	Official Syllabus Verification, Previous-Year Paper Sample & Question-Pattern Analysis	Mandatory Research
Part II	Evidence-Based Syllabus & Topic-Priority Map (Topics 1–21 Classified as Core / Supplementary)	Revision Strategy
Part III	Detailed Topic-Wise Notes & Solved Examples (Topics 1 to 21 — All Major Question Types in 8-Part A–H Format)	Core & Supp.
Part IV	Compact Formula & Shortcut Reference Tables (Tables A to H: Fractions, Powers, Speed, Work, 2D/3D, Traps)	Rapid Lookup
Part V	Research-Based Question-Pattern Observations, Distractor Design & Time-Management Strategy	Exam Analytics
Part VI	Final Mixed Practice Test (25 Exam-Style MCQs Across All Core Topics)	Self-Assessment
Part VII	Separate Answer Key & Complete Step-by-Step Worked Solutions for Mixed Practice Test	Verification
Part VIII	One-Page Rapid-Revision Formula Sheet, Final Exam-Day Checklist & Verified Source Links	Final Review

How to Use This Cheat Sheet: Start by reviewing the Evidence-Based Topic Priority Map in Part II so you allocate 70% of your study time to High-Priority topics. For every topic in Part III, study the 8-part template (A. Concept → B. Formula Box → C. Recognition Technique → D. Solved DSSSB MCQ → E. Step-by-Step Solution → F. Shortcut Method → G. Common Trap → H. Practice Variation). Use Part IV and Part VIII for final-week formula memorisation.

Part I: Mandatory Research Findings, Official Syllabus & PYQ Evidence

Before drafting any notes, official recruitment notifications and examination schemes published by the Delhi Subordinate Services Selection Board (https://dsssb.delhi.gov.in/) and archived shift papers from reputable educational repositories (Testbook DSSSB TGT Syllabus, Drishti Teaching Exams, BYJU'S DSSSB Official Tier-I Scheme) were examined. Below is the verbatim official syllabus for **Section A — (iii) Arithmetical & Numerical Ability**:

> *"Arithmetical & Numerical Ability: The test of Arithmetical and Numerical Abilities will cover Number Systems including questions on Simplification, Decimals, Data Interpretation, Fractions, L.C.M., H.C.F., Ratio & Proportion, Percentage, Average, Profit & Loss, Discount, Simple & Compound Interest, Mensuration, Time & Work, Time & Distance, Tables & Graphs, etc. of 10th level."*

1.1 Disclosed Sample of Verified DSSSB Previous-Year Papers Examined

Because DSSSB conducts Computer-Based Tests (CBTs) across multiple shifts and posts using a unified Section-A question bank, we examined **75 verified previous-year questions (PYQs)** drawn from five official DSSSB question compilations: (1) DSSSB TGT Arithmetic Ability Official Shifts (2021–2025), (2) DSSSB PRT Arithmetic Ability Official Shifts (2022–2025), (3) DSSSB LDC Official Shifts, (4) DSSSB Junior Assistant Official Shifts, and (5) DSSSB MTS & Warder Official Shifts. In addition, post-specific DSSSB TGT Mathematics Section-B questions were reviewed solely to distinguish Section-A general arithmetic from Section-B post-specific topics (such as Arithmetic Progressions, Trigonometry, and Statistics).
Methodological Disclosure: Findings below reflect the observed distribution across this accessible 75-question multi-paper sample and are indicative of DSSSB CBT patterns rather than a guarantee of future shift counts.

Part II: Research Summary Table & Evidence-Based Topic Priority Map

Topic & Syllabus Status	Verified Papers Examined	Observed Question Types in Verified Sample	Evidence-Based Priority
1. Number System *[Core Official Syllabus]*	DSSSB TGT (Q1, Q3, Q12, Q13), MTS (Q4), Warder (Q1)	6-digit divisibility by 36 (4 & 9 rule); total positive divisors of 8×81×125; factoring $7^{29}-7^{30}-7^{31}+7^{32}$; recurring decimal rationality; inclusion-exclusion divisibility (10 but not 3).	**High Priority** (6/75 sample Qs; 2–3 Qs per TGT shift)
2. Simplification & Approximation *[Core Official Syllabus]*	DSSSB TGT (Q10), PRT (Q1, Q6, Q10, Q11), LDC (Q10), Jr Asst (Q15), MTS (Q6, Q15)	Nested brackets with vinculum (bar); decimal algebraic identities a^3+b^3 and $a^3+b^3+c^3-3abc$; surd rationalization $(\sqrt{5}+\sqrt{3})/(\sqrt{5}-\sqrt{3})$; fraction numerator/denominator word problems.	**High Priority** (9/75 sample Qs; 2–3 Qs per TGT shift)
3. HCF and LCM *[Core Official Syllabus]*	DSSSB TGT (Q2, Q7), PRT (Q9), LDC (Q8), Jr Asst (Q13), MTS (Q9, Q11), Warder (Q7)	HCF of decimals (21.97, 15.21, 16.9, 18.59, 20.28); LCM with remainder 3 & exact divisibility by 67; smallest 6-digit number with remainder 2; largest divisor leaving same remainder (98, 190, 282, 466); least perfect square divisible by 5, 6, 10.	**High Priority** (8/75 sample Qs; 1–2 Qs per TGT shift)

Topic & Syllabus Status	Verified Papers Examined	Observed Question Types in Verified Sample	Evidence-Based Priority
4. Ratio and Proportion *[Core Official Syllabus]*	DSSSB PRT (Q5), LDC (Q11, Q12), Jr Asst (Q14), Warder (Q8, Q14)	4X = 6Y = 8Z share division; ratio balancing P:(Q+R)=3:4 and Q:(P+R)=2:3; coin denomination problems (Re 1, Rs 2, Rs 5 in 5:4:6); weighted headcount & per-person share ratios.	**High Priority** (6/75 sample Qs; 1–2 Qs per TGT shift)
5. Percentages *[Core Official Syllabus]*	DSSSB TGT (Q6), PRT (Q4, Q8, Q12), LDC (Q13), Jr Asst (Q6), MTS (Q8, Q10, Q12), Warder (Q10)	Price +20% vs Expenditure +10% consumption drop; price drop enabling +50 kg more rice for Rs 450; 4-person chained income percentages; successive changes & 2-year machine depreciation (9% p.a.); remaining income/apples problems.	**High Priority** (10/75 sample Qs; 2–3 Qs per TGT shift)
6. Profit, Loss & Discount *[Core Official Syllabus]*	DSSSB TGT (Q8), LDC (Q9), Jr Asst (Q5), Warder (Q11, Q13)	MP-CP-SP chain (10% discount on MP, 15% profit on CP, SP=Rs 2070); SP transition from 35% loss to 40% profit; articles per rupee (12 pens/Re 1 to 10 pens/Re 1); two-item zero net profit/loss.	**High Priority** (5/75 sample Qs; 1–2 Qs per TGT shift)
7. Simple & Compound Interest *[Core Official Syllabus]*	DSSSB TGT (Q9), LDC (Q5), Jr Asst (Q2, Q4), MTS (Q1, Q2, Q3, Q7), Warder (Q12)	Step-up SI (8% yr 1, +0.4%/yr for 4 yrs); half-yearly compounding (30% p.a. for 1 yr; 20% p.a. for 1.5 yrs); 2-yr CI vs SI difference (Rs 1040 vs Rs 1000); consecutive year CI ratio (Rs 3600 in 2 yrs, Rs 4320 in 3 yrs); tiered SI rates P% and Q%.	**High Priority** (9/75 sample Qs; 1–2 Qs per TGT shift)
8. Average *[Core Official Syllabus]*	DSSSB LDC (Q6), Jr Asst (Q3, Q8)	Overlapping group averages across 5 classes (A, B, C, D, E); multi-year brand average comparisons; replacement & inclusion/exclusion of observations.	**Medium Priority** (3/75 sample Qs; 1 Q per TGT shift)
9. Problems on Ages *[Core via Ratio & Linear Eq.]*	DSSSB MTS (Q13)	Past-to-future age ratio shift (6 yrs ago 7:8, 2 yrs hence 9:10 → sum of present ages); constant age difference method.	**Medium Priority** (1/75 sample Qs; 0–1 Q per shift)
10. Time and Work *[Core Official Syllabus]*	DSSSB LDC (Q2), Warder (Q3, Q5)	Fractional work rates (C does 2/10 in 4h, D does 2/5 in 5h → time for 13/20 work); work with daily hours (R1 6h×18d, R2 15h×12d → combined at 9h/day); $M_1D_1H_1 = M_2D_2H_2$.	**High Priority** (3/75 sample Qs; 1–2 Qs per TGT shift)
11. Pipes and Cisterns *[Core via Time & Work]*	DSSSB Jr Asst (Q10), Warder (Q15)	Two inlets (4.5h, 6h) + delayed outlet pipe (12h) opened after 1 hour; simultaneous two-pipe filling (48 min & 72 min).	**Medium Priority** (2/75 sample Qs; 0–1 Q per shift)
12. Time, Speed & Distance *[Core Official Syllabus]*	DSSSB LDC (Q4, Q7), Jr Asst (Q7), Warder (Q4)	4 equal distances (40 km each) at 10, 20, 30, 40 km/h (harmonic mean 19.2 km/h); inverse speed-time ratio before/after repair; opposite relative speed of car & man + bridge crossing.	**High Priority** (4/75 sample Qs; 1–2 Qs per TGT shift)
13. Trains *[Core via Speed & Distance]*	DSSSB LDC (Q15), Jr Asst (Q11), Warder (Q9)	Two trains (600 m & 1000 m) crossing in opposite directions in 96 s; staggered departure times (6:30 pm & 7:30 pm) meeting at 11:30 pm → distance ratio	**High Priority** (3/75 sample Qs; 1 Q per TGT shift)

Topic & Syllabus Status	Verified Papers Examined	Observed Question Types in Verified Sample	Evidence-Based Priority
		EG:FG; opposite trains meeting after 19 hours.	
14. Boats and Streams *[Core Extension of TSD]*	Comparable DSSSB One-Tier shifts	Upstream/downstream speed transformation, still-water & stream speed calculation, and round-trip travel time equations.	**Medium Priority** (Rotates with Trains in TSD slot)
15. Mixture and Alligation *[Core Extension of Ratio/%]*	DSSSB Warder (Q6)	Ingredient addition to flip mixture ratio (45 L sugar:salt 1:2 → 2:1); alligation cross-rule for weighted averages; repeated liquid replacement formula.	**Medium Priority** (1/75 sample Qs; 0–1 Q per shift)
16. Unitary Method & Work-Wages *[Core Official Syllabus]*	DSSSB Warder (Q2, Q3, Q14)	Direct packing/cost division (8064 apples in 32 boxes); inverse manpower-days; wages distributed strictly in ratio of work contributed.	**Medium Priority** (3/75 sample Qs; integrated with Work/Ratio)
17. Mensuration: 2D Shapes *[Core Official Syllabus]*	DSSSB TGT (Q5), PRT (Q7, Q15), MTS (Q5, Q14)	Quadrilateral area via Pythagorean triplet (9, 40, 41) + Heron's formula (15, 28, 41); regular hexagon area (side 6 cm = 54√3); circle radius −40% area drop (−64%); parallelogram base/altitude; circumradius of isosceles triangle (12, 9, 9).	**High Priority** (5/75 sample Qs; 1–2 Qs per TGT shift)
18. Mensuration: 3D Shapes *[Core Official Syllabus]*	DSSSB LDC (Q3, Q14), Jr Asst (Q1, Q12), JE (Q8)	Max cylinder cut from 3×4×5 cuboid (264/7 cm^3); hemisphere volume & CSA statement check (r=3.5 cm); cube TSA vs volume statement check (a=8 cm); volume ratio of cone:sphere:cube (22:1617:72).	**High Priority** (5/75 sample Qs; 1–2 Qs per TGT shift)
19. Basic Algebra & Equations *[Core via Simplification/Number]*	DSSSB TGT (Q14), PRT (Q2)	Two-variable linear word problem (cows & hens: 92 legs, 29 heads → 17 cows); polynomial factor theorem (x−3 factor of $x^2+4xp-3p$ → p = −1); algebraic identities.	**Medium Priority** (2/75 sample Qs; 1 Q per shift)
20. Data Interpretation (Tables & Graphs) *[Core Official Syllabus]*	DSSSB TGT (Q4, Q11), PRT (Q3, Q13, Q14), LDC (Q1, Q6), Jr Asst (Q8, Q9)	Multi-bar chart composite expressions ($J_2-J_1+J_3$); Income-Expenditure-Savings bar chart % comparison; 7-candidate election table ratio; 5-pulse & 7-procedure pie charts; multi-statement table verification.	**High Priority** (9/75 sample Qs; 2–3 Qs per TGT shift)
21. Additional Relevant Topics *[Supplementary / Post-Dependent]*	DSSSB TGT (Q15), TGT Maths (Q1, Q12), Adda247 TGT PDF (Q37)	Arithmetic Progressions (AP nth term & sum of 51 terms); elementary Trigonometric identity simplification; basic Mean/Median/Mode & Permutations/Probability (primarily TGT Maths Section-B, occasionally 1 Q in general shifts).	**Supplementary** (Post-dependent; revise after Core 1–20)

Part III: Detailed Topic-Wise Notes & Verified Solved Examples (Topics 1–21)

Every topic below is classified according to the verified official DSSSB Section-A syllabus and includes complete coverage of all subtopics plus worked solutions following the mandatory 8-part pedagogical template (A. Concept, B. Formula Box, C. Recognition Technique, D. Solved DSSSB MCQ, E. Step-by-Step Solution, F. Shortcut Method, G. Common Trap, H. Practice Variation).

Topic 1: Number System [Core Syllabus — High Priority]

Subtopics Covered: Natural (1, 2, 3...), Whole (0, 1, 2...), Integers (...−2, −1, 0, 1, 2...), Rational (p/q where q ≠ 0, including terminating and non-terminating recurring decimals), and Irrational numbers (√2, √3, π — non-terminating non-recurring); Prime (exactly 2 factors; 2 is the only even prime; 1 is neither prime nor composite; 25 primes from 1–100; 15 primes from 1–50) and Composite numbers; Odd and Even parity rules; Factors/Divisors and Multiples; Divisibility rules (2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 13, 36, 72, 88, 99); Unit digit cyclicity; Remainder theorem & exponential divisibility; Consecutive integers & Digit-interchange problems.

1.1. Composite Divisibility Rules (Divisibility by 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 13, 36, 72, 88, 99)

A. Concept (Class-10 Foundation): To test if a large number is divisible by a composite divisor D (such as 36, 72, 88, or 99), split D into two co-prime factors (a, b) such that HCF(a, b) = 1 (e.g., 36 = 4 × 9; 72 = 8 × 9; 88 = 8 × 11; 99 = 9 × 11). The number must satisfy both individual rules simultaneously.

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Divisibility by 2, 4, 8:** Last 1 digit (for 2), last 2 digits (for 4), or last 3 digits (for 8) must be divisible by 2, 4, or 8 respectively. • **Divisibility by 3 & 9:** Sum of all digits must be divisible by 3 or 9. • **Divisibility by 5 & 10:** Last digit is 0 or 5 (for 5); last digit is 0 (for 10). • **Divisibility by 11:** |(Sum of digits at odd places) − (Sum of digits at even places)| = 0 or a multiple of 11. • **Divisibility by 7 & 13:** Group digits in triplets from right to left; |(Sum of odd triplets) − (Sum of even triplets)| is divisible by 7 or 13.	Look for a 5- to 9-digit number with one or two missing digits (e.g., '58579A' or '45792B') stated to be 'exactly divisible by 36, 72, 88, or 99'.

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB TGT Arithmetic Ability (Question 1)]:
If the 6-digit number 58579A is exactly divisible by 36, then what is the value of the digit A?
(A) 2 **(B)** 4 **(C)** 6 **(D)** 8

E. Step-by-Step Solution:

1. **Given Information:** 6-digit number = 58579A (where A is the units digit, 0 ≤ A ≤ 9); Divisor = 36.
2. **Required Quantity:** The single digit value of A.
3. **Formula / Method:** Factor 36 into co-prime factors 4 and 9 (since HCF(4, 9) = 1). Apply the divisibility rule of 9 (sum of digits divisible by 9) and the rule of 4 (last two digits '9A' divisible by 4).
4. **Substitution of Values:** Sum of digits = 5 + 8 + 5 + 7 + 9 + A = 34 + A. Last two digits = 9A.
5. **Calculation:** For 34 + A to be divisible by 9, the next multiple of 9 after 34 is 36, so 34 + A = 36 ⇒ A = 2. Check divisibility by 4: when A = 2, the last two digits form 92, and 92 ÷ 4 = 23 (exact). Thus A = 2 satisfies both 9 and 4.
6. **Final Answer: 2 (Option A)**

F. Shortcut Method (Speed Technique): Digital Root Elimination: Cancel digits summing to 9 in '58579A' → cancel 9; 5+5+8 = 18 (multiple of 9, cancel); left with 7 + A. For 7 + A = 9, A must be 2 in 3 seconds!

G. Common Trap & Distractor Alert: Factoring 36 as 6 × 6 or 12 × 3 (not co-prime) or checking only divisibility by 4 (which gives A = 2 or A = 6; Option C = 6 is a trap for students who skip the 9-rule).

H. Additional Practice Variation & Verified Solution:
Question: If the 6-digit number 45792B is exactly divisible by 72, find the value of B. Also, in DSSSB MTS (Q4), how many numbers are there between 501 and 701 which are divisible by 10 but NOT divisible by 3?
Verified Solution: (1) For 45792B ÷ 72: Co-prime factors are 8 and 9. Sum of digits = 4+5+7+9+2+B = 27+B ⇒ B = 0 or 9. Last 3 digits '92B' divisible by 8: 920 ÷ 8 = 115 (exact), whereas 929 ÷ 8 leaves remainder 1. So **B = 0**. (2) Between 501 and 701: Multiples of 10 are 510, 520, ..., 700 → (700−510)/10 + 1 = 20 numbers. Multiples of LCM(10,3)=30 are 510, 540, ..., 690 → (690−510)/30 + 1 = 7 numbers. Required count = 20 − 7 = **13 numbers**.

1.2. Prime Factorisation, Total Positive Divisors, Odd/Even Factors & Prime Properties

A. Concept (Class-10 Foundation): Every composite integer N > 1 can be uniquely expressed as a product of prime powers: N = 2^p × a^q × b^r..., where 2 is the even prime and a, b are distinct odd primes. Any divisor is formed by choosing an exponent from 0 to p for 2, 0 to q for a, and so on.

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
If **$N = 2^p \times a^q \times b^r$** (where a, b are odd primes): • **Total Positive Divisors (Factors):** (p + 1)(q + 1)(r + 1) • **Odd Divisors (no factor of 2):** (q + 1)(r + 1) • **Even Divisors (at least one 2):** p × (q + 1)(r + 1) • **Sum of All Divisors:** $[(2^{p+1}-1)/(2-1)] \times [(a^{q+1}-1)/(a-1)] \times [(b^{r+1}-1)/(b-1)]$ • **Rational vs Irrational:** Any non-terminating recurring decimal (e.g., 0.555... = 5/9) is ALWAYS a **rational number** (Verified DSSSB TGT Q13).	Phrases like 'Find the number of positive divisors/factors of...', 'How many even/odd factors does N have?', or 'Sum of squares of prime numbers between X and Y'.

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB TGT Arithmetic Ability (Question 12)]:
The number of positive divisors of 8 × 81 × 125 is:
(A) 20 **(B)** 80 **(C)** 16 **(D)** 100

E. Step-by-Step Solution:

1. **Given Information:** Product N = 8 × 81 × 125.
2. **Required Quantity:** Total number of positive divisors (factors) of N.
3. **Formula / Method:** Convert each composite number into its prime base power, combine like prime bases, and multiply (exponent + 1) for each distinct prime.
4. **Substitution of Values:** 8 = 2^3; 81 = 3^4; 125 = 5^3. Thus N = 2^3 × 3^4 × 5^3 (with p = 3, q = 4, r = 3).
5. **Calculation:** Total number of positive divisors = (3 + 1) × (4 + 1) × (3 + 1) = 4 × 5 × 4 = 80.
6. **Final Answer: 80 divisors (Option B)**

F. Shortcut Method (Speed Technique): Spot instantly that $8=2^3$, $81=3^4$, $125=5^3$ are already coprime prime powers. Add 1 to each power: 4 × 5 × 4 = 80.

G. Common Trap & Distractor Alert: Multiplying the exponents directly without adding 1 (3 × 4 × 3 = 36) or failing to reduce composite bases (like 8, 81, 125) to prime bases first.

H. Additional Practice Variation & Verified Solution:
Question: (a) [Verified PYQ — DSSSB Warder Q1] Find the sum of squares of prime numbers between 20 and 30. (b) Find the number of even divisors of 360.
Verified Solution: (a) Primes between 20 and 30 are 23 and 29 (note: 21=3×7, 27=3^3 are composite). Sum of squares = $23^2 + 29^2$ = 529 + 841 = **1370**. (b) 360 = 36 × 10 = $2^3 \times 3^2 \times 5^1$. Number of even divisors = p(q+1)(r+1) = 3 × (2+1) × (1+1) = 3 × 3 × 2 = **18**.

1.3. Exponential Divisibility, Remainder Theorem, Unit Digit Cyclicity & Consecutive/Digit Problems

A. Concept (Class-10 Foundation): When large powers are added or subtracted (e.g., a^n ± a^(n+1) ± a^(n+2)), factor out the smallest power a^n. For unit digits of a^b, only the unit digit of base 'a' and the exponent 'b mod 4' matter because all digits repeat their unit digit every 4th power.

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Exponential Factoring:** $a^n + a^{n+1} + a^{n+2} = a^n(1 + a + a^2)$ • **Algebraic Divisibility:** $(a^n - b^n)$ is always divisible by (a − b), and also by (a + b) when n is even. $(a^n + b^n)$ is divisible by (a + b) when n is odd. • **Unit Digit Cyclicity:** Digits **0, 1, 5, 6** always end in themselves; **4, 9** have cycle 2 (4^{odd}=4, 4^{even}=6; 9^{odd}=9, 9^{even}=1); **2, 3, 7, 8** have cycle 4 (divide exponent by 4; if remainder r=0, use power 4). • **Two-Digit Number (10x + y):** Reversing digits gives (10y + x). Difference = $\lvert 9(x - y) \rvert$ (always divisible by 9); Sum = 11(x + y) (always divisible by 11).	Expressions with high exponents like '7^29 - 7^30 - 7^31 + 7^32 is divisible by...', 'Find the unit digit of...', or 'Remainder when 17^200 is divided by 18'.

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB TGT Arithmetic Ability (Question 3)]:
Which of the following numbers will completely divide (7^29 − 7^30 − 7^31 + 7^32)?
(A) 26 **(B)** 78 **(C)** 72 **(D)** 51

E. Step-by-Step Solution:

1. **Given Information:** Exponential expression E = 7^29 − 7^30 − 7^31 + 7^32.
2. **Required Quantity:** Which option (26, 78, 72, or 51) exactly divides E.
3. **Formula / Method:** Factor out the smallest power 7^29 from all four terms and evaluate the resulting integer bracket.
4. **Substitution of Values:** E = 7^29 × (7^0 − 7^1 − 7^2 + 7^3) = 7^29 × (1 − 7 − 49 + 343).
5. **Calculation:** Inside the bracket: 1 − 7 − 49 + 343 = 344 − 56 = 288. So E = 7^29 × 288. Check which option divides 288: 288 ÷ 72 = 4 (exact integer), whereas 26, 78, and 51 do not divide 288 or 7^29.
6. **Final Answer: 72 (Option C)**

F. Shortcut Method (Speed Technique): Factor by grouping pairs: 7^29(1 − 7) − 7^31(1 − 7) = (1 − 7)(7^29 − 7^31) = (−6) × 7^29 × (1 − 49) = (−6)(−48) × 7^29 = 288 × 7^29 = 72 × 4 × 7^29.

G. Common Trap & Distractor Alert: Making a sign error inside the bracket (writing 1 − 7 − 49 − 343) or trying to use remainder cyclicity on each option instead of factoring 7^29.

H. Additional Practice Variation & Verified Solution:
Question: (a) Find the unit digit of (23)^45 × (57)^98 × (78)^63. (b) If a two-digit number and the number formed by reversing its digits differ by 54, find the difference between its digits.
Verified Solution: (a) Take unit digits of bases and exponents mod 4: For 3^45, 45 mod 4 = 1 ⇒ 3^1 = 3. For

7^98, 98 mod 4 = 2 ⇒ $7^2 = 49$ (ends in 9). For 8^63, 63 mod 4 = 3 ⇒ $8^3 = 512$ (ends in 2). Product of unit digits = 3 × 9 × 2 = 54 ⇒ Unit digit is **4**. (b) Difference $|(10x+y) - (10y+x)| = 9|x - y| = 54 \Rightarrow |x - y| = 54 \div 9 = \mathbf{6}$.

Topic 2: Simplification and Approximation [Core Syllabus — High Priority]

Subtopics Covered: VBODMAS / PEMDAS hierarchy (Vinculum/Bar → Brackets `()`, `{}`, `[]` → Of → Division & Multiplication left-to-right → Addition & Subtraction left-to-right); Fractions & Mixed fractions; Decimal operations & Recurring decimals to fractions; Algebraic identities in decimal simplification; Squares, cubes, roots, indices, laws of exponents, and surd rationalization; Approximation & fast mental calculation.

2.1. VBODMAS Hierarchy, Nested Brackets & Vinculum (Bar) Operations

A. Concept (Class-10 Foundation): In any arithmetic expression, operations must follow the strict VBODMAS order. Crucially: (1) A Vinculum (horizontal bar over an expression like 16 − 48) acts as an innermost bracket and must be evaluated BEFORE the minus sign outside it; (2) Multiplication and Division have EQUAL priority and must be executed strictly from LEFT to RIGHT (unless 'of' is written, which binds tighter than ÷); (3) Addition and Subtraction also have EQUAL priority and are evaluated from LEFT to RIGHT.

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **V (Vinculum / Bar):** Evaluate expression directly under the bar first: − a − b = −(a − b). • **B (Brackets):** Innermost to outermost: Round ( ) → Curly { } → Square [ ]. • **O (Of / Orders):** 'a ÷ b of c' means a ÷ (b × c), whereas 'a ÷ b × c' means (a ÷ b) × c. • **D & M (Left-to-Right):** Division and Multiplication at the same level. • **A & S (Left-to-Right):** Addition and Subtraction at the same level.	Expressions with nested brackets [{ ( ) }] and a horizontal bar (vinculum) over the last two numbers, or chains of ÷ and × using the same number (like 12 ÷ 12 × 12).

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB LDC Arithmetic Ability (Question 10)]:
What is the value of $48 - [48 - (16 + 48) - 48 - (48 - \overline{16 - 48}) + 32]$?
(A) 160 **(B)** 144 **(C)** 96 **(D)** 64

E. Step-by-Step Solution:

1. **Given Information:** Expression = $48 - [48 - (16 + 48) - 48 - (48 - \overline{16 - 48}) + 32]$.
2. **Required Quantity:** Exact simplified numerical value.
3. **Formula / Method:** Evaluate the vinculum (bar) over 16 − 48 first, then round brackets ( ), then square brackets [ ], and finally subtract from the outer 48.
4. **Substitution of Values:** Step 1 (Vinculum): $\overline{16 - 48} = -32$. Substitute into its round bracket: (48 − (−32)) = 48 + 32 = 80.
5. **Calculation:** Step 2 (First round bracket): (16 + 48) = 64. Step 3 (Square bracket): [48 − 64 − 48 − 80 + 32]. Notice 48 − 48 = 0, leaving [−64 − 80 + 32] = −112. Step 4 (Outer subtraction): 48 − [−112] = 48 + 112 = 160.
6. **Final Answer: 160 (Option A)**

F. Shortcut Method (Speed Technique): Cancel 48 − 48 = 0 inside the square bracket immediately, evaluate the bar 16−48 = −32 so −(48 − (−32)) = −80, and sum: 48 − (−64 − 80 + 32) = 48 + 112 = 160.

G. Common Trap & Distractor Alert: Treating $-\overline{16-48}$ as $-16-48=-64$ instead of $-(16-48)=+32$. That sign mistake gives $48-(48-64)=64$ or 96 (Options C and D).

H. Additional Practice Variation & Verified Solution:
Question: (a) [Verified PYQ — DSSSB Jr. Assistant Q15] Find the value of $1728-[12\times\{12\div(12\div\overline{12\times12})\}]$. (b) [Verified PYQ — DSSSB PRT Q1] Simplify: $25\div5+3\times60-(17\times200\div40)$.
Verified Solution: (a) Under vinculum: $\overline{12\times12}=144$. Round bracket: $(12\div144)=1/12$. Curly bracket: $\{12\div(1/12)\}=12\times12=144$. Square bracket: $[12\times144]=1728$. Final: $1728-1728=$ **0**. (b) Bracket first: $(17\times200\div40)=17\times5=85$. Then $25\div5=5$ and $3\times60=180$. Final: $5+180-85=$ **100**.

2.2. Decimal Algebraic Identities & Recurring Decimal Conversions

A. Concept (Class-10 Foundation): DSSSB frequently disguises simple algebraic formulas like a^3+b^3, a^3-b^3, $(a-b)^2$, or $a^3+b^3+c^3-3abc$ using messy decimals (like 0.86, 0.14, 1.75, 0.75). Never compute the cubes or squares manually! Instead, match the decimal expression to the standard identity and track the decimal shift (powers of 10) between numerator and denominator.

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Sum of Cubes:** $(a^3+b^3)/(a^2-ab+b^2)=$ **a + b** • **Difference of Cubes:** $(a^3-b^3)/(a^2+ab+b^2)=$ **a – b** • **Three-Term Cubic Identity:** $(a^3+b^3+c^3-3abc)/(a^2+b^2+c^2-ab-bc-ca)=$ **a + b + c** • **Square Identities:** $a^2\pm2ab+b^2=(a\pm b)^2$; $(a+b)^2+(a-b)^2=2(a^2+b^2)$; $(a+b)^2-(a-b)^2=4ab$ • **Recurring Decimals to Fractions:** $0.\overline{ab}=ab/99$; $0.\overline{abc}=(abc-a)/990$ (one 9 per repeating digit, one 0 per non-repeating decimal digit).	Three or two cubed decimals in the numerator (e.g., $0.86^3+0.14^3$) and squared decimals in the denominator (e.g., $8.6^2+1.4^2-86\times0.14$) where the base digits are identical.

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB TGT Arithmetic Ability (Question 10)]:
The value of $[(0.86)^3+(0.14)^3]\div[(8.6)^2+(1.4)^2-86\times0.14]$ is equal to:
(A) 10 **(B)** 0.01 **(C)** 0.001 **(D)** 0.1

E. Step-by-Step Solution:

1. **Given Information:** Numerator $N=(0.86)^3+(0.14)^3$; Denominator $D=(8.6)^2+(1.4)^2-86\times0.14$.
2. **Required Quantity:** Exact decimal value of $N\div D$.
3. **Formula / Method:** Let $a=0.86$ and $b=0.14$. Express the denominator in terms of a and b by factoring out $10^2=100$, then apply $(a^3+b^3)/(a^2+b^2-ab)=a+b$.
4. **Substitution of Values:** Notice $8.6=10a$, $1.4=10b$, and $86\times0.14=(100a)(b)=100ab$. Thus $D=(10a)^2+(10b)^2-100ab=100(a^2+b^2-ab)$.
5. **Calculation:** $N\div D=[a^3+b^3]\div[100(a^2+b^2-ab)]=(a+b)\div100$. Since $a+b=0.86+0.14=1.00$, the value is $1\div100=0.01$.
6. **Final Answer: 0.01 (Option B)**

F. Shortcut Method (Speed Technique): Ignore decimals first: (a+b) gives 0.86 + 0.14 = 1. Now check decimal scaling: each term in the denominator is shifted 1 decimal place right and squared, meaning D is $10^2=100$ times larger. So 1 / 100 = 0.01.

G. Common Trap & Distractor Alert: Assuming $N/(a^2+b^2-ab)=a+b=1$ without noticing that the denominator uses 8.6^2 instead of 0.86^2 (which traps students into picking 0.1 or 1).

H. Additional Practice Variation & Verified Solution:
Question: (a) [Verified PYQ — DSSSB PRT Q10] Find $[(2.5)^3 + (4.7)^3 + (2.8)^3 - 7.5 \times 4.7 \times 2.8] \div [25^2 + 47^2 + 28^2 - 25\times47 - 47\times28 - 28\times25]$. (b) [Verified PYQ — DSSSB PRT Q6] Evaluate $1.75 \times 1.75 - 3.5 \times 0.75 + 0.75 \times 0.75$.
Verified Solution: (a) Let a=2.5, b=4.7, c=2.8 (note 3a = 7.5). Numerator = $a^3+b^3+c^3-3abc = (a+b+c)(a^2+b^2+c^2-ab-bc-ca)$. Denominator has 25=10a, 47=10b, 28=10c, so D = $100(a^2+b^2+c^2-ab-bc-ca)$. Ratio = (a+b+c)/100 = (2.5+4.7+2.8)/100 = 10/100 = **0.1**. (b) Recognize $a^2 - 2ab + b^2$ with a=1.75, b=0.75 (since 2×1.75 = 3.5). Value = $(1.75 - 0.75)^2 = 1^2$ = **1**.

2.3. Surds, Indices, Rationalization & Fraction Word Problems

A. Concept (Class-10 Foundation): To simplify a fraction with a surd denominator like $(\sqrt{x} + \sqrt{y})/(\sqrt{x} - \sqrt{y})$, multiply both numerator and denominator by the conjugate of the denominator, $(\sqrt{x} + \sqrt{y})$, using $(\sqrt{x} - \sqrt{y})(\sqrt{x} + \sqrt{y}) = x - y$. For fraction word problems, set the original fraction as N/D and form linear equations from the given numerator/denominator adjustments.

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Conjugate Rationalization Shortcut:** $(\sqrt{x} + \sqrt{y}) / (\sqrt{x} - \sqrt{y})$ = **$[(x + y) + 2\sqrt{(xy)}] / (x - y)$** • **Sum of Reciprocal Conjugates:** $(\sqrt{x} + \sqrt{y})/(\sqrt{x} - \sqrt{y}) + (\sqrt{x} - \sqrt{y})/(\sqrt{x} + \sqrt{y})$ = **$2(x + y) / (x - y)$** • **Laws of Indices:** $a^m \times a^n = a^{m+n}$; $a^m \div a^n = a^{m-n}$; $(a^m)^n = a^{mn}$; $a^{-n} = 1/a^n$; $a^0 = 1$ $(a \neq 0)$.	Equations equating a surd fraction to 'a + b√k' and asking for 'a + b', or word problems stating 'The numerator of a fraction is k more than its denominator...'.

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB PRT Arithmetic Ability (Question 11)]:
If $(\sqrt{5} + \sqrt{3}) \div (\sqrt{5} - \sqrt{3}) = a + b\sqrt{15}$, then what is the value of (a + b)?
(A) 10 **(B)** 8 **(C)** 4 **(D)** 5

E. Step-by-Step Solution:

1. **Given Information:** $(\sqrt{5} + \sqrt{3}) / (\sqrt{5} - \sqrt{3}) = a + b\sqrt{15}$, where a and b are rational numbers.
2. **Required Quantity:** The sum (a + b).
3. **Formula / Method:** Rationalize the denominator by multiplying numerator and denominator by $(\sqrt{5} + \sqrt{3})$, then equate rational and irrational parts.
4. **Substitution of Values:** LHS = $(\sqrt{5} + \sqrt{3})^2 \div [(\sqrt{5})^2 - (\sqrt{3})^2] = (5 + 3 + 2\sqrt{15}) \div (5 - 3)$.
5. **Calculation:** LHS = $(8 + 2\sqrt{15}) \div 2 = 4 + 1\sqrt{15}$. Comparing $4 + 1\sqrt{15}$ with $a + b\sqrt{15}$ gives a = 4 and b = 1. Therefore, a + b = 4 + 1 = 5.
6. **Final Answer: 5 (Option D)**

F. Shortcut Method (Speed Technique): When x − y = 2 (as in 5 − 3 = 2), $(\sqrt{x} + \sqrt{y})/(\sqrt{x} - \sqrt{y})$ always simplifies directly to $[(x+y)/2] + \sqrt{(xy)}$. Here (5+3)/2 = 4 and coefficient of √15 is 1, so 4 + 1 = 5 in 5 seconds!

G. Common Trap & Distractor Alert: Forgetting to divide the 2√15 term by the denominator 2 (writing a = 4, b = 2 and choosing a + b = 6 or 8).

H. Additional Practice Variation & Verified Solution:
Question: [Verified PYQ — DSSSB MTS Q15] The numerator of a fraction is 3 more than the denominator. When 5 is added to the numerator and 2 is subtracted from the denominator, the fraction becomes 8/3. When the original fraction is divided by 5½, the fraction obtained is:
Verified Solution: Let denominator = d, numerator = d + 3. Given: $(d + 3 + 5)/(d - 2) = 8/3 \Rightarrow 3(d + 8) = 8(d -$

2) ⇒ 3d + 24 = 8d − 16 ⇒ 5d = 40 ⇒ d = 8. Original fraction = 11/8. Dividing by 5½ (= 11/2): (11/8) ÷ (11/2) = (11/8) × (2/11) = 2/8 = **1/4**.

Topic 3: HCF and LCM [Core Syllabus — High Priority]

Subtopics Covered: Prime factorisation & Euclid's division method; Product relation (HCF × LCM = $N_1 \times N_2$) & Co-prime representation ($N_1 = hx$, $N_2 = hy$ with HCF(x,y)=1); HCF & LCM of fractions and decimals; Remainder-based LCM problems (same remainder, constant difference, and additional divisibility condition); Remainder-based HCF problems (leaving given or equal remainders); Bells, traffic lights, and circular track recurring intervals; Least perfect square divisible by a set of numbers.

3.1. HCF & LCM of Decimals and Fractions, Product Relation & Co-Prime Pairs

A. Concept (Class-10 Foundation): To find the HCF or LCM of decimal numbers, first equalize the number of decimal places across all numbers (by appending trailing zeros), multiply by 10^k to turn them into integers, find the integer HCF or LCM, and divide back by 10^k. For two numbers with HCF = h, always write the numbers as hx and hy where x and y are co-prime (HCF(x, y) = 1); then LCM = hxy.

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **HCF of Fractions:** HCF(Numerators) ÷ LCM(Denominators) • **LCM of Fractions:** LCM(Numerators) ÷ HCF(Denominators) • **Two-Number Product Rule:** $N_1 \times N_2$ = HCF × LCM • **Co-Prime Representation:** If $N_1 = hx$ and $N_2 = hy$ with HCF(x, y) = 1, then $N_1 + N_2 = h(x + y)$ and LCM = hxy. • **Sum of Reciprocals:** $1/N_1 + 1/N_2 = (N_1 + N_2) / (HCF \times LCM)$.	Decimals with 1 or 2 decimal places asking for HCF/LCM, or two-number problems giving HCF, LCM, and sum/difference.

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB TGT Arithmetic Ability (Question 2)]:
What is the Highest Common Factor (HCF) of 21.97, 15.21, 16.9, 18.59 and 20.28?
(A) 1.3 **(B)** 1.69 **(C)** 13 **(D)** 3

E. Step-by-Step Solution:

1. **Given Information:** Five decimal numbers: 21.97, 15.21, 16.90, 18.59, and 20.28.
2. **Required Quantity:** The Highest Common Factor (HCF) of the five numbers.
3. **Formula / Method:** Equalize to 2 decimal places (write 16.9 as 16.90), multiply all by 100 to obtain integers, use the difference method (HCF must divide the difference between consecutive sorted integers), and divide by 100.
4. **Substitution of Values:** Integers (×100) sorted: 1521, 1690, 1859, 2028, 2197.
5. **Calculation:** Differences between adjacent terms: 1690 − 1521 = 169; 1859 − 1690 = 169; 2028 − 1859 = 169; 2197 − 2028 = 169. Check if 169 divides 1521: 1521 = 169 × 9. Indeed, the numbers are 169×9, 169×10, 169×11, 169×12, 169×13. Since HCF(9,10,11,12,13) = 1, the integer HCF is 169. Dividing by 100 gives 1.69.
6. **Final Answer: 1.69 (Option B)**

F. Shortcut Method (Speed Technique): Difference Method for HCF: The HCF of any set of numbers ALWAYS divides the smallest difference between any two of them! Here 16.90 − 15.21 = 1.69, and Option B is 1.69.

G. Common Trap & Distractor Alert: Forgetting to write 16.9 as 16.90 (treating it as 169 instead of 1690), which falsely yields 13 or 1.3 (Options A and C).

H. Additional Practice Variation & Verified Solution:
Question: (a) [Verified PYQ — DSSSB PRT Q9] The HCF and LCM of two numbers are 36 and 144 respectively. Find the sum of their reciprocals. (b) [Verified PYQ — DSSSB MTS Q11] The sum and difference of LCM and HCF of two numbers are 638 and 580, and the sum of the two numbers is 290. Find the numbers.
Verified Solution: (a) Let numbers be 36x and 36y with HCF(x,y)=1. Then 36xy = 144 ⇒ xy = 4. Co-prime pair for xy=4 is (1, 4), so numbers are 36 and 144. Sum of reciprocals = 1/36 + 1/144 = **5/144**. (b) LCM = (638+580)/2 = 609; HCF = (638−580)/2 = 29. Numbers are 29a and 29b: 29(a+b)=290 ⇒ a+b=10, and ab = 609/29 = 21 ⇒ {a,b} = {3, 7}. Numbers are 29×3 = **87** and 29×7 = **203**.

3.2. Remainder-Based LCM & HCF Problems, Least Perfect Square & Interval Bells

A. Concept (Class-10 Foundation): When a number leaves the same remainder r when divided by d_1, d_2, d_3, its general form is N = k × LCM(d_1, d_2, d_3) + r. If there is no extra divisibility condition, set k = 1. If N must also be divisible by another divisor M (like 67), test k = 1, 2, 3... using modular remainders until (LCM × k + r) is divisible by M.

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Type 1 (Same Remainder r, Divisible by M):** N = k × LCM(d_1, d_2, ...) + r; find smallest integer k ≥ 1 such that M divides N. • **Type 2 (Different Remainders with Constant Gap c = $d_1-r_1 = d_2-r_2$):** N = k × LCM(d_1, d_2, ...) − c. • **Type 3 (Greatest Divisor Leaving Same Unknown Remainder on a, b, c):** Divisor = **HCF(|b − a|, |c − b|, |c − a|)**. • **Type 4 (Least Perfect Square Divisible by d_1, d_2, ...):** Prime-factorize LCM and raise every odd prime exponent to the next even number.	Phrases like 'leaves a remainder of r in each case, but is exactly divisible by M', 'smallest 6-digit number leaving remainder r', or 'largest number which divides a, b, c leaving the same remainder'.

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB TGT Arithmetic Ability (Question 7)]:
Let x be the least number which when divided by 21, 33, 35 and 55 leaves a remainder of 3 in each case, but is exactly divisible by 67. What is the sum of the digits of x?
(A) 12 **(B)** 15 **(C)** 17 **(D)** 16

E. Step-by-Step Solution:

1. **Given Information:** Divisors = 21, 33, 35, 55; common remainder r = 3; exact divisor M = 67.
2. **Required Quantity:** Sum of the digits of the least such number x.
3. **Formula / Method:** Compute L = LCM(21, 33, 35, 55), write x = L·k + 3, reduce L modulo 67 to find k quickly, and sum the digits of x.
4. **Substitution of Values:** 21 = 3×7; 33 = 3×11; 35 = 5×7; 55 = 5×11. Thus L = LCM = 3 × 5 × 7 × 11 = 1155. General form: x = 1155k + 3.
5. **Calculation:** Divide 1155 by 67: 1155 = 67 × 17 + 16. So (1155k + 3) mod 67 = (16k + 3) mod 67. Test k = 1, 2, 3, 4: For k = 4, 16(4) + 3 = 64 + 3 = 67, which is divisible by 67! Thus x = 1155 × 4 + 3 = 4620 + 3 = 4623. Sum of digits of 4623 = 4 + 6 + 2 + 3 = 15.
6. **Final Answer: 15 (Option B)**

F. Shortcut Method (Speed Technique): Why does DSSSB ask for 'sum of digits of x'? So you cannot back-substitute options! Always reduce 1155 mod 67 = 16 first; solving 16k + 3 = 67 ⇒ 16k = 64 ⇒ k = 4 takes 5 seconds instead of multiplying 1155×1, 1155×2, 1155×3.

G. Common Trap & Distractor Alert: Stopping at k = 1 (x = 1158, digit sum = 15 by coincidence here, but in most shifts k = 1 gives a wrong digit sum) or forgetting to add the remainder +3 at the end.

H. Additional Practice Variation & Verified Solution:
Question: (a) [Verified PYQ — DSSSB LDC Q8] Find the smallest 6-digit number which when divided by 3, 5, 7, 9 and 15 leaves remainder 2 in each case. (b) [Verified PYQ — DSSSB Jr. Asst Q13] Find the largest number which divides 98, 190, 282 and 466 leaving the same remainder in each case. (c) [Verified PYQ — DSSSB MTS Q9] Find the least square number divisible by 5, 6, and 10.
Verified Solution: (a) LCM(3,5,7,9,15) = 315. Divide 100000 by 315: 100000 = 315×317 + 145. Smallest 6-digit multiple of 315 = 100000 + (315−145) = 100170. Add remainder 2: **100172**. (b) Required divisor = HCF of differences = HCF(190−98, 282−190, 466−282) = HCF(92, 92, 184) = **92**. (c) LCM(5,6,10) = 30 = $2^1 \times 3^1 \times 5^1$. Make all powers even: $2^2 \times 3^2 \times 5^2 = 30^2$ = **900**.

Topic 4: Ratio and Proportion [Core Syllabus — High Priority]

Subtopics Covered: Simplifying ratios & Equivalent ratios; Direct and Inverse proportion; Dividing a quantity in a given ratio; Combining ratios (A:B and B:C) & Equal-multiple relations (pA = qB = rC); Ratio balancing when total sum is invariant; Fourth, Third, and Mean proportional (Continued proportion); Coin denomination & Headcount-share problems; Partnership investment-time ratios.

4.1. Ratio Balancing (Total Sum Invariance), Combining Ratios & Equal-Multiple Division

A. Concept (Class-10 Foundation): When the same total money (P + Q + R) is split in two different ratio statements — such as P : (Q + R) = a : b and Q : (P + R) = c : d — the total number of ratio units (a + b) and (c + d) must represent the exact same total sum. Equalize the total units by multiplying each ratio so both sums equal LCM(a + b, c + d).

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Total-Sum Balancing:** If P : (Q+R) = a : b (sum = S_1) and Q : (P+R) = c : d (sum = S_2), then P's fraction of total = **a / (a+b)** and Q's fraction = **c / (c+d)**. • **Equal-Multiple Rule (pX = qY = rZ):** Divide by LCM(p, q, r) → **X : Y : Z = (1/p) : (1/q) : (1/r) = qr : pr : pq**. • **Continued Proportion (a : b :: b : c):** Mean proportional b = √(ac); Third proportional c = b^2/a; Fourth proportional of a, b, c is d = bc/a.	Problems stating 'P's share is 3/4 of the total share of Q and R, and Q's share is 2/3 of the total share of P and R' or '4 times X's share equals 6 times Y's share equals 8 times Z's share'.

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB LDC Arithmetic Ability (Question 12)]:
Rs. 70,000 is divided among P, Q and R such that P's share is 3/4 of the total share of Q and R, and Q's share is 2/3 of the total share of P and R. What is the share of R?
(A) Rs. 16,000 **(B)** Rs. 18,000 **(C)** Rs. 14,000 **(D)** Rs. 12,000

E. Step-by-Step Solution:

1. **Given Information:** Total sum (P + Q + R) = Rs. 70,000; P : (Q + R) = 3 : 4; Q : (P + R) = 2 : 3.
2. **Required Quantity:** The share of R (in Rs.).
3. **Formula / Method:** Equalize total units of (P + Q + R) across both ratios using LCM(3+4, 2+3) = LCM(7, 5) = 35 units.

4. **Substitution of Values:** In Ratio 1, total = 3 + 4 = 7 units; multiply by 5 ⇒ P : (Q + R) = 15 : 20. In Ratio 2, total = 2 + 3 = 5 units; multiply by 7 ⇒ Q : (P + R) = 14 : 21.
5. **Calculation:** Out of 35 total units, P = 15 units and Q = 14 units, so R = 35 − (15 + 14) = 6 units. Since 35 units = Rs. 70,000 ⇒ 1 unit = Rs. 2,000. Therefore, R's share = 6 × 2,000 = Rs. 12,000.
6. **Final Answer: Rs. 12,000 (Option D)**

F. Shortcut Method (Speed Technique): Direct Fraction Method: P = (3/7)×70,000 = 30,000; Q = (2/5)×70,000 = 28,000; R = 70,000 − 58,000 = Rs. 12,000.

G. Common Trap & Distractor Alert: Taking P = 3/4 of 70,000 instead of 3/(3+4) = 3/7 of 70,000! When P : (Q+R) = 3 : 4, P is 3/7 of the total, NOT 3/4.

H. Additional Practice Variation & Verified Solution:
Question: [Verified PYQ — DSSSB PRT Q5] Rs. 7,800 is divided among X, Y and Z in such a way that 4 times X's share is equal to 6 times Y's share which is equal to 8 times Z's share. What is the share of Y?
Verified Solution: Given 4X = 6Y = 8Z. Divide through by LCM(4, 6, 8) = 24: X/6 = Y/4 = Z/3 ⇒ X : Y : Z = 6 : 4 : 3. Total units = 6 + 4 + 3 = 13 units. Since 13 units = Rs. 7,800 ⇒ 1 unit = Rs. 600. Share of Y = 4 units = 4 × 600 = **Rs. 2,400**.

4.2. Coin Denomination Value Ratios, Group Headcount Ratios & Partnership

A. Concept (Class-10 Foundation): In coin problems, always distinguish between the 'ratio of the NUMBER of coins' ($n_1 : n_2 : n_3$) and the 'ratio of their MONETARY VALUE' ($v_1n_1 : v_2n_2 : v_3n_3$). Similarly, in company profit-sharing or partnership problems, total group share = (number of persons in group) × (share per person), and partnership profit ratio = ($Capital_1 \times Time_1$) : ($Capital_2 \times Time_2$).

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Total Value from Coin Count Ratio:** If coins of denominations d_1, d_2, d_3 are in count ratio $n_1 : n_2 : n_3$, then Total Value = $\mathbf{(d_1n_1 + d_2n_2 + d_3n_3)x}$ while Total Coins = $\mathbf{(n_1 + n_2 + n_3)x}$. • **Partnership Profit Split:** $Profit_A : Profit_B = (Investment_A \times Months_A) : (Investment_B \times Months_B)$.	Questions mentioning coin denominations (Re 1, Rs 2, Rs 5) with total coins or total value, or group profit ratios across managers, supervisors, and clerks.

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB Warder Arithmetic Ability (Question 8)]:
A child gets 1 random coin out of Re 1, Rs 2, and Rs 5 coins daily. He starts on 1st April and at the end of the month, he has coins in the ratio 5 : 4 : 6. Find the total value of the coins with him.
(A) Rs. 84 **(B)** Rs. 94 **(C)** Rs. 96 **(D)** Rs. 86

E. Step-by-Step Solution:

1. **Given Information:** Denominations = Re 1, Rs 2, Rs 5; Ratio of number of coins = 5 : 4 : 6; Total coins = 1 coin/day for the month of April = 30 coins.
2. **Required Quantity:** Total monetary value of the 30 coins (in Rs.).
3. **Formula / Method:** Equate the sum of coin-count ratio units (5x + 4x + 6x) to the 30 days in April to find x, then multiply each coin count by its rupee denomination.
4. **Substitution of Values:** 5x + 4x + 6x = 30 ⇒ 15x = 30 ⇒ x = 2.
5. **Calculation:** Number of Re 1 coins = 5 × 2 = 10 (Value = 10 × 1 = Rs. 10). Number of Rs 2 coins = 4 × 2 = 8 (Value = 8 × 2 = Rs. 16). Number of Rs 5 coins = 6 × 2 = 12 (Value = 12 × 5 = Rs. 60). Total value = 10 + 16 + 60 = Rs. 86.

6. **Final Answer: Rs. 86 (Option D)**

F. Shortcut Method (Speed Technique): Value of 1 basic ratio set (5, 4, 6 → 15 coins) = 5(1) + 4(2) + 6(5) = 5 + 8 + 30 = Rs. 43. Since April has 30 days (2 sets of 15 coins), Total Value = 2 × 43 = Rs. 86!

G. Common Trap & Distractor Alert: Assuming April has 31 days instead of 30 days, or multiplying the total value by the count ratio directly.

H. Additional Practice Variation & Verified Solution:
Question: [Verified PYQ — DSSSB Warder Q14] A company shares total profit among managers, supervisors and clerks in the ratio 4 : 3 : 2. There are 8 managers, 12 supervisors and 20 clerks. If each manager gets Rs. 25,000, find the sum of the amount received by one supervisor and one clerk.
Verified Solution: Total share of 8 managers = 8 × 25,000 = Rs. 2,00,000, which corresponds to 4 ratio units ⇒ 1 unit = Rs. 50,000. Total share of 12 supervisors (3 units) = Rs. 1,50,000 ⇒ 1 supervisor gets 1,50,000 ÷ 12 = Rs. 12,500. Total share of 20 clerks (2 units) = Rs. 1,00,000 ⇒ 1 clerk gets 1,00,000 ÷ 20 = Rs. 5,000. Sum = 12,500 + 5,000 = **Rs. 17,500**.

Topic 5: Percentages [Core Syllabus — High Priority]

Subtopics Covered: Fraction-decimal-percentage conversions; Percentage increase/decrease & Reference base shifts ('A is what % more/less than B'); Product constancy (Expenditure = Price × Consumption) & Quantity-price reduction problems; Successive percentage changes, Chained comparisons & Machine depreciation; Income, expenditure, and savings problems; Election & Examination pass/fail reverse percentage problems.

5.1. Product Constancy (Expenditure = Price × Consumption) & Price-Drop Extra Quantity

A. Concept (Class-10 Foundation): Whenever Expenditure = Price × Consumption (or Area = Length × Breadth), a percentage change in Price and Expenditure determines the exact percentage change in Consumption via Consumption = Expenditure ÷ Price. When a price reduction of r% allows a buyer to purchase ΔQ kg more for a fixed outlay of Rs. E, the money saved on the original purchase (r% of E) pays for the extra ΔQ kg at the NEW (reduced) price.

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Expenditure-Price-Consumption Ratio:** $C_2 / C_1 = (E_2 / E_1) \div (P_2 / P_1)$ • **Zero Expenditure Change (Price ±r%):** Consumption change = **$[\mp r / (100 \pm r)] \times 100\%$** • **Price Reduction Extra Quantity (Rs. E buys ΔQ more when price drops r%):** – **New (Reduced) Price per kg:** $(r / 100) \times (E / \Delta Q)$ – **Original Price per kg:** $[r / (100 - r)] \times (E / \Delta Q)$	Problems stating 'Price of oil increases by x%, expenditure increases by only y%, find % change in consumption' or 'A reduction of r% in price enables a person to buy ΔQ kg more for Rs. E'.

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB TGT Arithmetic Ability (Question 6)]:
The price of oil is increased by 20%. However, the expenditure on it increases by 10% only. What is the percentage increase or decrease in consumption?
(A) 8⅓% increase **(B)** 8⅓% decrease **(C)** 10% decrease **(D)** 10% increase

E. Step-by-Step Solution:

1. **Given Information:** Price increase = +20% (Price ratio $P_1 : P_2 = 100 : 120 = 5 : 6$); Expenditure increase = +10% (Expenditure ratio $E_1 : E_2 = 100 : 110 = 10 : 11$).
2. **Required Quantity:** Percentage increase or decrease in oil consumption.

3. **Formula / Method:** Set initial Price = 100, initial Expenditure = 100 (or use Consumption = Expenditure / Price).
4. **Substitution of Values:** New Price matches 120; New Expenditure matches 110. Thus Consumption changes from 120 units to 110 units (since 120 acts as the break-even expenditure for constant consumption).
5. **Calculation:** Decrease in consumption = (120 − 110) ÷ 120 × 100% = (10 / 120) × 100% = (1 / 12) × 100% = 8⅓% decrease.
6. **Final Answer: 8⅓% decrease (Option B)**

F. Shortcut Method (Speed Technique): Base-120 Bridge: Start at 100. Price goes to 120, Expenditure goes to 110. To go from 120 back to 110, you must drop by 10 on a base of 120 → −10/120 = −1/12 = −8⅓% in 3 seconds!

G. Common Trap & Distractor Alert: Subtracting percentages directly (20% − 10% = 10% decrease, Option C) or dividing by 100 instead of the new price 120.

H. Additional Practice Variation & Verified Solution:
Question: [Verified PYQ — DSSSB MTS Q8] The reduction of 20% in the price of rice enables a person to obtain 50 kg more for Rs. 450. Find the original price of rice per kg.
Verified Solution: Method 1 (Ratio): Price ratio Before : After = 5 : 4 ⇒ Consumption ratio for fixed Rs. 450 = 4 : 5. Difference = 1 unit = 50 kg ⇒ Original consumption (4 units) = 200 kg. Original price = Rs. 450 ÷ 200 kg = **Rs. 2.25/kg**. Method 2 (Formula): Original Price = [20/(100−20)] × (450/50) = (1/4) × 9 = **Rs. 2.25/kg**.

5.2. Successive Percentage Changes, Chained Comparisons & Income/Remainder Problems

A. Concept (Class-10 Foundation): When a quantity undergoes two successive percentage changes of a% and b% (such as a 40% increase followed by a 30% decrease, or a 9% annual depreciation for 2 years), the second percentage acts on the already changed value, NOT just the original value. For chained comparisons across 3 or 4 people (A → B → C → D), multiply the fractional multipliers in sequence.

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Net Successive Percentage Change (2 steps): [a + b + (ab / 100)]%** (use + for increase, − for decrease). • **Fractional Multiplier Chain:** Final = Initial × (1 ± r_1/100) × (1 ± r_2/100) × ... • **2-Year Depreciation at r% p.a.:** Total % Decrease = **(2r − r^2/100)%**. • **Base Comparison:** If A and B are x% and y% less than C (so C=100, A=100−x, B=100−y), then % by which A exceeds B = **[(A − B) / B] × 100%**.	Problems with 'increased by a% and then decreased by b%', 'depreciates at r% per annum for 2 years', or 'A is 60% more than B, B is 75% more than C, C is 20% less than D'.

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB Junior Assistant Arithmetic Ability (Question 6)]:
A's income is 60% more than B's income. B's income is 75% more than C's income and C's income is 20% less than D's income. If the income of D is Rs. 20,000, then what is the income of A?
(A) Rs. 13,200 **(B)** Rs. 11,200 **(C)** Rs. 44,800 **(D)** Rs. 22,400

E. Step-by-Step Solution:

1. **Given Information:** D = Rs. 20,000; C = 20% less than D; B = 75% more than C; A = 60% more than B.
2. **Required Quantity:** The income of A (in Rs.).

3. **Formula / Method:** Convert each percentage change into a simplified fraction multiplier and multiply D by all three multipliers.
4. **Substitution of Values:** −20% → 1 − 1/5 = 4/5; +75% → 1 + 3/4 = 7/4; +60% → 1 + 3/5 = 8/5. Thus A = D × (4/5) × (7/4) × (8/5).
5. **Calculation:** Cancel 4 in the numerator and denominator: A = 20,000 × (7 × 8) ÷ (5 × 5) = 20,000 × (56 / 25) = 800 × 56 = Rs. 44,800.
6. **Final Answer: Rs. 44,800 (Option C)**

F. Shortcut Method (Speed Technique): Fraction cancellation (4/5 × 7/4 × 8/5 = 56/25) eliminates intermediate rupee calculations completely!

G. Common Trap & Distractor Alert: Adding the percentages linearly (−20% + 75% + 60% = +115% → 20,000 × 2.15 = 43,000) instead of compounding the multipliers.

H. Additional Practice Variation & Verified Solution:
Question: (a) [Verified PYQ — DSSSB PRT Q12] A machine worth Rs. 1,800 depreciates at 9% p.a. Find the percentage decrease in its value after 2 years. (b) [Verified PYQ — DSSSB PRT Q8] Two numbers are less than a third number by 28% and 36% respectively. By what percent is the first number greater than the second? (c) [Verified PYQ — DSSSB MTS Q10] A fruit seller has apples of which 5% are rotten. He sells 75% of the remainder and is left with 190 apples. Find the initial number of apples.
Verified Solution: (a) Ignore Rs. 1,800! For two successive −9% drops, Net % decrease = 9 + 9 − (9×9)/100 = 18 − 0.81 = **17.19%**. (b) Let 3rd number = 100 ⇒ 1st = 72, 2nd = 64. Required % = [(72−64)/64] × 100 = (8/64)×100 = **12.5%**. (c) Remaining fraction = (95/100) × (25/100) × N = 190 ⇒ (19/20) × (1/4) × N = 190 ⇒ N = 10 × 80 = **800 apples**.

Topic 6: Profit, Loss and Discount [Core Syllabus — High Priority]

Subtopics Covered & Base Principle: Cost Price (CP), Selling Price (SP), and Marked/List Price (MP); **Crucial Base Rule:** Profit and Loss percentages are ALWAYS calculated on **Cost Price (CP)** (because profit/loss measures return on what the seller spent), whereas Discount percentage is ALWAYS calculated on **Marked Price (MP)** (because a discount is a reduction off the printed tag price); Single & successive discounts; MP-to-CP direct ratio; Articles bought/sold per rupee; Dishonest shopkeeper / false-weight problems.

6.1. Marked Price, Cost Price & Selling Price Chain (Discount + Profit Together)

A. Concept (Class-10 Foundation): When a shopkeeper or dealer offers/receives a discount of d% on Marked Price (MP) and still earns a profit of p% on Cost Price (CP), the Selling Price (SP) bridges both prices: SP = MP × (100 − d)/100 = CP × (100 + p)/100. If a dealer BUYS at a d% discount on MP, then the dealer's Cost Price is CP = MP × (100 − d)/100.

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Profit % (always on CP):** [(SP − CP) / CP] × 100% ⇒ SP = CP × (100 + p)/100 • **Loss % (always on CP):** [(CP − SP) / CP] × 100% ⇒ SP = CP × (100 − l)/100 • **Discount % (always on MP):** [(MP − SP) / MP] × 100% ⇒ SP = MP × (100 − d)/100 • **Direct MP : CP Ratio (when seller gives d% discount and earns p% profit): MP / CP = (100 + p) / (100 − d)** • **Two Equivalent Successive Discounts (d_1%, d_2%):** Single Discount = **$[d_1 + d_2 - (d_1 d_2)/100]$%**.	Problems linking Marked Price, Discount %, Profit %, and Selling Price, or asking for the new SP to convert an l% loss into a p% profit.

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB TGT Arithmetic Ability (Question 8)]:
A dealer buys articles at a discount of 10% on their marked price and sells them at 15% profit. If he sells the goods for Rs. 2,070, then the marked price of the articles for the dealer is:
(A) Rs. 1,600 **(B)** Rs. 1,800 **(C)** Rs. 2,000 **(D)** Rs. 2,400

E. Step-by-Step Solution:

1. **Given Information:** Dealer's purchase discount on MP = 10%; Dealer's profit on his CP = 15%; Dealer's Selling Price (SP) = Rs. 2,070.
2. **Required Quantity:** The Marked Price (MP) of the articles.
3. **Formula / Method:** Express the dealer's CP as 90% of MP, then express his SP as 115% of his CP, and equate to Rs. 2,070.
4. **Substitution of Values:** Dealer's CP = MP × (90 / 100). Dealer's SP = CP × (115 / 100) = MP × (90 / 100) × (115 / 100) = 2,070.
5. **Calculation:** First find CP: CP × (115 / 100) = 2,070 ⇒ CP = (2,070 × 100) ÷ 115 = 18 × 100 = Rs. 1,800. Next find MP: MP × (9 / 10) = 1,800 ⇒ MP = (1,800 × 10) ÷ 9 = Rs. 2,000.
6. **Final Answer: Rs. 2,000 (Option C)**

F. Shortcut Method (Speed Technique): Assume MP = 100 units ⇒ Dealer's CP = 90 units ⇒ Dealer's SP = 90 + 15% of 90 = 90 + 13.5 = 103.5 units. Since 103.5 units = Rs. 2,070 ⇒ 1 unit = 20 ⇒ MP (100 units) = Rs. 2,000!

G. Common Trap & Distractor Alert: Stopping after finding the Cost Price Rs. 1,800 (which is Option B — a classic DSSSB distractor!) or applying MP/CP = (100+15)/(100−10) blindly when the discount was on buying rather than selling.

H. Additional Practice Variation & Verified Solution:
Question: [Verified PYQ — DSSSB LDC Q9] A man loses 35% on selling 45 pens for Rs. 130. What should be the selling price of 45 pens to earn a profit of 40%?
Verified Solution: Ignore '45 pens' since the quantity is identical in both cases! 35% loss means SP_1 = 65% of CP = Rs. 130 ⇒ 1% of CP = 130 ÷ 65 = Rs. 2. For 40% profit, SP_2 = 140% of CP = 140 × 2 = **Rs. 280**.

6.2. Articles Per Rupee, Equated CP/SP Multiples & Dishonest Shopkeeper (False Weight)

A. Concept (Class-10 Foundation): For a fixed rupee amount (like Re. 1), the number of articles is inversely proportional to the price per article. Thus, to change from Price Factor $F_1 = (100 \pm r_1)\%$ to Price Factor $F_2 = (100 \pm r_2)\%$, the number of articles per rupee changes in the inverse ratio $F_1 : F_2$. In false-weight problems, a dishonest seller who gives W grams instead of 1000 grams has a true cost of W and a revenue of 1000 (adjusted for any markup/discount).

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Articles Per Rupee Rule:** $(Articles_1 \times PriceFactor_1) / Rupees_1 = (Articles_2 \times PriceFactor_2) / Rupees_2$ • **Equated Multiples (m × CP = n × SP):** SP / CP = m / n ⇒ Profit/Loss % = **[(m − n) / n] × 100%** • **Dishonest Dealer (False Weight W g instead of 1000 g at Cost Price):** Profit % = **[(1000 − W) / W] × 100%** • **Two Items Sold at Same SP (one at +x% profit, one at −x% loss):** Always results in a net **Loss of $(x^2 / 100)$%**.	Questions stating 'Bought 12 pens for a rupee, how many should he sell for a rupee to gain 20%?', '3 times SP equals 2 times CP', or 'uses 800g weight instead of 1 kg'.

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB Junior Assistant Arithmetic Ability (Question 5)]:
A man bought 12 pens for a rupee (cost price of all pens is equal). How many pens should he sell for a rupee to gain 20 percent?
(A) 5 **(B)** 6 **(C)** 10 **(D)** 8

E. Step-by-Step Solution:

1. **Given Information:** CP of 12 pens = Re. 1 (Price Factor F_1 = 100%); Target gain = +20% (Price Factor F_2 = 120%).
2. **Required Quantity:** Number of pens N_2 to sell for Re. 1.
3. **Formula / Method:** Use inverse proportionality between price per pen (5 : 6) and quantity of pens per rupee (6 : 5).
4. **Substitution of Values:** 12 × 100 = N_2 × 120.
5. **Calculation:** N_2 = (12 × 100) ÷ 120 = 1,200 ÷ 120 = 10 pens.
6. **Final Answer: 10 pens (Option C)**

F. Shortcut Method (Speed Technique): Price increases by 20% (ratio 5 : 6), so quantity per rupee must be in ratio 6 : 5. Since 6 units = 12 pens ⇒ 5 units = 10 pens instantly!

G. Common Trap & Distractor Alert: Multiplying 12 by 1.2 to get 14.4 or dividing by 2 (picking 6). To earn a profit for the SAME 1 rupee, you must give FEWER pens, not more!

H. Additional Practice Variation & Verified Solution:
Question: (a) [Verified PYQ — DSSSB Warder Q11] If three times the SP of an article is equal to two times its CP, find the profit or loss percentage. (b) A dishonest grocer promises to sell at cost price but uses a weight of 800 grams for 1 kg. Find his profit percentage.
Verified Solution: (a) Given 3 × SP = 2 × CP ⇒ SP / CP = 2 / 3. Since SP < CP, there is a loss of 1 unit on a CP of 3 units: Loss % = (1/3) × 100% = **33.33%**. (b) True cost = 800 g; Revenue = 1000 g. Profit % = [(1000 − 800) / 800] × 100% = (200 / 800) × 100% = **25%**.

Topic 7: Simple and Compound Interest [Core Syllabus — High Priority]

Subtopics Covered: Principal (P), Rate (R% p.a.), Time (T), Simple Interest (SI), Compound Interest (CI), and Amount (A); Step-up/variable rate SI & Tiered investment rates; Money doubling/tripling under SI vs CI; Annual, half-yearly, and quarterly compounding; Difference between CI and SI for 2 and 3 years; Consecutive-year CI amount ratios.

7.1. Simple Interest: Step-Up Rates, Tiered Slabs & Money Multiplication

A. Concept (Class-10 Foundation): In Simple Interest, interest is calculated strictly on the original Principal P every year. Therefore, if the annual interest rate changes across years (R_1%, R_2%, R_3%...), the total Simple Interest is simply (R_1 + R_2 + R_3 + ...)% of P. Similarly, if a sum becomes n times itself in T years under SI, the interest earned is (n − 1)P.

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Standard Simple Interest:** SI = (P × R × T) / 100; Amount A = P + SI = P[1 + (RT/100)] • **Variable / Step-Up Annual Rates:** Total SI = P × **(R_1 + R_2 + ... + R_n) / 100** • **Money Multiplication under SI:** If a sum becomes n_1	Questions where the SI rate increases by a fixed step each year (e.g., '8% in first year with an increase of 0.4% in each subsequent year'), or 'doubles in 12 years, triples in how many years?'.

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
times in T_1 years and n_2 times in T_2 years: **$(n_1 - 1) / T_1 = (n_2 - 1) / T_2$**	

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB TGT Arithmetic Ability (Question 9)]:
A person takes a loan at a simple interest of 8% per annum for the first year with an increase of 0.4% in each subsequent year. If he paid Rs. 8,428 as interest after 4 years, how much loan did he take?
(A) Rs. 23,800 **(B)** Rs. 24,200 **(C)** Rs. 24,500 **(D)** Rs. 24,000

E. Step-by-Step Solution:

1. **Given Information:** Rates for 4 years: R_1 = 8%, R_2 = 8.4%, R_3 = 8.8%, R_4 = 9.2%; Total SI after 4 years = Rs. 8,428.
2. **Required Quantity:** Principal loan amount P (in Rs.).
3. **Formula / Method:** Sum the 4 annual percentage rates to get the cumulative effective SI percentage, then divide Total SI by that percentage.
4. **Substitution of Values:** Cumulative Rate R_total = 8 + 8.4 + 8.8 + 9.2 = 34.4%. Thus 34.4% of P = 8,428.
5. **Calculation:** P = (8,428 × 100) ÷ 34.4 = 8,42,800 ÷ 344. Divide by 8: 1,05,350 ÷ 43 = Rs. 24,500.
6. **Final Answer: Rs. 24,500 (Option C)**

F. Shortcut Method (Speed Technique): Notice 344 × 2 = 688 and 842 − 688 = 154; then 1548 ÷ 344 = 4.5. So 8428 ÷ 34.4 = 245 ⇒ P = Rs. 24,500.

G. Common Trap & Distractor Alert: Using R = 8% for all 4 years (32%) or applying compound interest instead of simple rate addition.

H. Additional Practice Variation & Verified Solution:
Question: (a) [Verified PYQ — DSSSB MTS Q7] A sum of money doubles itself with simple interest in 12 years. In how many years would it triple itself? (b) [Verified PYQ — DSSSB Jr. Asst Q4] A man earns P% on the first Rs. 3,000 and Q% on the rest. He earns Rs. 750 from Rs. 6,000 and Rs. 1,050 from Rs. 8,000. Find P.
Verified Solution: (a) Using $(n_1-1)/T_1 = (n_2-1)/T_2$: $(2-1)/12 = (3-1)/T_2 \Rightarrow 1/12 = 2/T_2 \Rightarrow T_2$ = **24 years**. (b) Extra investment = 8,000 − 6,000 = Rs. 2,000 earns extra interest 1,050 − 750 = Rs. 300 ⇒ Q% = (300/2,000)×100 = 15%. On Rs. 6,000, the second Rs. 3,000 at 15% gives Rs. 450, so the first Rs. 3,000 earns 750 − 450 = Rs. 300 ⇒ P = (300/3,000)×100 = **10%**.

7.2. Compound Interest: Half-Yearly/Quarterly Compounding, CI−SI Difference & Consecutive Years

A. Concept (Class-10 Foundation): Under Compound Interest, interest earned in each period is added to the principal for the next period. For half-yearly compounding, divide the annual rate by 2 (r = R/2) and double the years (n = 2T); for quarterly compounding, r = R/4 and n = 4T. For 2 years, the difference (CI − SI) is simply the interest on the first year's SI: CI − SI = r% of (1st year SI) = $P(r/100)^2$.

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Compound Amount & CI:** $A = P(1 + r/100)^n$; $CI = A - P$ • **Half-Yearly:** r = R/2, n = 2T | **Quarterly:** r = R/4, n = 4T • **2-Period CI vs SI Difference:** $CI_2 - SI_2$ = **$P(R/100)^2$** = **$(R / 200) \times SI_2$** • **3-Period CI vs SI Difference:** $CI_3 - SI_3$ = **$P(R/100)^2(3 +$**	Mentions of 'compounded half-yearly', 'simple and compound interest earned in two years is Rs. X and Rs. Y', or 'becomes Rs. A_1 in 2 years and Rs. A_2 in 3 years'.

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
R/100) • **Consecutive Amounts (A_t and A_{t+1}):** Rate R% = **$[(A_{t+1} - A_t) / A_t] \times 100\%$**.	

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB LDC Arithmetic Ability (Question 5)]:
A man borrowed Rs. 12,000 at compound interest at the rate of 30 percent per annum. If the interest is compounded half-yearly, then what will be the amount to be paid after 1 year?
(A) Rs. 15,870 **(B)** Rs. 16,240 **(C)** Rs. 15,940 **(D)** Rs. 15,880

E. Step-by-Step Solution:

1. **Given Information:** Principal P = Rs. 12,000; Annual rate R = 30% p.a.; Time T = 1 year; Compounded half-yearly.
2. **Required Quantity:** Total Amount A to be paid after 1 year.
3. **Formula / Method:** Convert to half-yearly parameters: r = 30% ÷ 2 = 15% per half-year, and n = 1 × 2 = 2 half-years. Compute $A = P(1 + 15/100)^2$.
4. **Substitution of Values:** $A = 12,000 \times (1 + 3/20)^2 = 12,000 \times (23/20) \times (23/20)$.
5. **Calculation:** $A = 12,000 \times (529 / 400) = 30 \times 529 = Rs. 15,870$.
6. **Final Answer: Rs. 15,870 (Option A)**

F. Shortcut Method (Speed Technique): Effective 2-period rate for 15%: 15 + 15 + (15×15)/100 = 32.25%. Amount = 132.25% of 12,000 = 120 × 132.25 = Rs. 15,870.

G. Common Trap & Distractor Alert: Calculating simple interest 30% of 12,000 = 3,600 (giving 15,600) or forgetting to halve the rate to 15%.

H. Additional Practice Variation & Verified Solution:
Question: (a) [Verified PYQ — DSSSB Warder Q12] The SI and CI earned in 2 years at the same rate on a certain sum are Rs. 1,000 and Rs. 1,040 respectively. Find the annual rate of interest. (b) [Verified PYQ — DSSSB Jr. Asst Q2] An amount becomes Rs. 3,600 in 2 years and Rs. 4,320 in 3 years under annual CI. If double the principal is invested for 3 years, what amount is obtained?
Verified Solution: (a) 1-year SI = 1,000 ÷ 2 = Rs. 500. Extra CI in 2nd year = 1,040 − 1,000 = Rs. 40, which is interest on the first year's Rs. 500: Rate R = (40/500) × 100% = **8% p.a.** (b) Since Amount is directly proportional to Principal ($A = P(1+r/100)^3$), doubling the Principal for the same 3 years simply doubles the 3-year amount: 2 × 4,320 = **Rs. 8,640** (no need to even find P or R!).

Topic 8: Average [Core Syllabus — Medium/High Priority]

Subtopics Covered: Basic arithmetic average & Total sum (Sum = Average × Count); Overlapping group averages; Adding or removing an observation (Teacher/Captain joining or leaving); Replacement of a member or batting/bowling average changes; Weighted averages across multiple sections; Average of consecutive natural, even, odd, or arithmetic progression numbers.

8.1. Overlapping Group Averages, Inclusion/Exclusion & Replacement Formulas

A. Concept (Class-10 Foundation): Whenever the average of a group of N items is A, immediately convert it to its Total Sum = N × A. In multi-group problems (like classes A, B, C, D, E), subtracting the sum of (A + C + D) from (A + B + C + D) isolates the exact value of B. When a new member replaces an old member and shifts the average of N people by ±ΔA, the new member's value differs from the old member's value by ±(N × ΔA).

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Fundamental Sum Rule:** Total Sum = Average × Number of Observations • **Weighted Average (Groups n_1, n_2 with averages A_1, A_2):** $A_{combined} = (n_1A_1 + n_2A_2) / (n_1 + n_2)$ • **Member Replacement (Count N stays constant):** New Value = **Old Value ± (N × ΔAverage)** • **Member Joining (Count becomes N + 1):** New Member = **Old Average ± [(N + 1) × ΔAverage]** • **Consecutive Numbers / AP:** Average = (First Term + Last Term) / 2 = Middle Term.	Problems listing averages of overlapping subsets (e.g., 'A and B is 39; A, B, C is 42; A, B, C, D is 45...') or 'When a teacher joins, average age increases by 1 year'.

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB Junior Assistant Arithmetic Ability (Question 3)]:
The average number of students in class A and B is 39; in class A, B and C is 42; in class A, B, C and D is 45; in class A, C and D is 46; and in class B and E is 45. What is the average number of students per class considering all five classes together?
(A) 45.6 **(B)** 44.8 **(C)** 42.2 **(D)** 40.4

E. Step-by-Step Solution:

1. **Given Information:** A+B = 39×2 = 78; A+B+C = 42×3 = 126; A+B+C+D = 45×4 = 180; A+C+D = 46×3 = 138; B+E = 45×2 = 90.
2. **Required Quantity:** Average of all 5 classes: (A + B + C + D + E) ÷ 5.
3. **Formula / Method:** Notice we already know (A + B + C + D) = 180! We only need E. Find B by subtracting (A + C + D) from (A + B + C + D), then find E from (B + E).
4. **Substitution of Values:** B = (A + B + C + D) − (A + C + D) = 180 − 138 = 42. Substitute B = 42 into B + E = 90 ⇒ E = 90 − 42 = 48.
5. **Calculation:** Total students in all 5 classes = (A + B + C + D) + E = 180 + 48 = 228. Average of 5 classes = 228 ÷ 5 = 45.6.
6. **Final Answer: 45.6 (Option A)**

F. Shortcut Method (Speed Technique): Skip A+B and A+B+C completely! They are superfluous data designed to waste time. You only need (A+B+C+D) = 180, (A+C+D) = 138 → B = 42 → E = 90 − 42 = 48 → (180+48)/5 = 45.6!

G. Common Trap & Distractor Alert: Wasting 2 minutes finding A, B, C, D individually from all 5 equations instead of spotting that (A+B+C+D) + E only requires E.

H. Additional Practice Variation & Verified Solution:
Question: The average weight of 15 students in a group increases by 1.5 kg when a student weighing 40 kg is replaced by a new student. What is the weight of the new student?
Verified Solution: Using the Replacement Formula: New Weight = Old Weight + (N × ΔAverage) = 40 + (15 × 1.5) = 40 + 22.5 = **62.5 kg**.

Topic 9: Problems on Ages [Core Syllabus — Medium Priority]

Subtopics Covered: Present, past ('t years ago'), and future ('t years hence') ages; Constant age difference principle; Balancing two age ratios across a time gap; Sum and average of family ages over time.

9.1. Age Ratio Shifts Across Past & Future (Constant Age Difference Method)

A. Concept (Class-10 Foundation): The most powerful law in age problems is that the difference between the ages of any two people remains CONSTANT throughout their lives (if A is 4 years older than B today, A was 4 years older 6 years ago and will be 4 years older 10 years from now). Therefore, when given two age ratios at two different points in time, equalize the difference between the ratio parts in both ratios; then the increase in each person's ratio units directly equals the total time gap!

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Time Gap Rule:** Gap between 't_1 years ago' and 't_2 years hence' = **$t_1 + t_2$ years**. • **Balanced Ratio Shortcut:** If Age Ratio at time 1 is a : b and at time 2 is c : d with (b − a) = (d − c), then: **1 Ratio Unit = (Total Time Gap) / (c − a) years**. • **If (b − a) ≠ (d − c):** Multiply Ratio 1 by $\lvert d - c \rvert$ and Ratio 2 by $\lvert b - a \rvert$ to make the age difference equal first. • **Sum of Ages Shift:** For n people, Sum of ages after t years = Present Sum + **n × t**.	Word problems giving the ratio of ages 'x years ago' and 'y years from now' and asking for present ages or sum of present ages.

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB MTS Arithmetic Ability (Question 13)]:
Six years ago, the ratio of the ages of A and B was 7 : 8. After 2 years from now, the ratio of their ages will be 9 : 10. What is the sum of the present ages (in years) of A and B?
(A) 57 **(B)** 102 **(C)** 72 **(D)** 87

E. Step-by-Step Solution:

1. **Given Information:** Ratio 6 years ago = 7 : 8; Ratio 2 years hence = 9 : 10; Total time gap = 6 + 2 = 8 years.
2. **Required Quantity:** Sum of present ages of A and B (in years).
3. **Formula / Method:** Check the horizontal difference in both ratios: 8 − 7 = 1 unit and 10 − 9 = 1 unit (already balanced!). Equate the vertical shift (9 − 7 = 2 units) to the 8-year time gap.
4. **Substitution of Values:** 2 ratio units = 8 years ⇒ 1 ratio unit = 4 years.
5. **Calculation:** Ages 6 years ago: A = 7 × 4 = 28 years, B = 8 × 4 = 32 years (Sum 6 years ago = 60 years). Present ages (add 6 years to each): A = 28 + 6 = 34 years, B = 32 + 6 = 38 years. Sum of present ages = 34 + 38 = 72 years.
6. **Final Answer: 72 years (Option C)**

F. Shortcut Method (Speed Technique): Sum 6 years ago = (7 + 8) × 4 = 60. For 2 people over 6 years, add 2 × 6 = 12 ⇒ Present Sum = 60 + 12 = 72 years!

G. Common Trap & Distractor Alert: Adding only 6 instead of 2 × 6 = 12 to the past sum (60 + 6 = 66), or taking the time gap as 6 − 2 = 4 years instead of 6 + 2 = 8 years.

H. Additional Practice Variation & Verified Solution:
Question: Four years ago, the ratio of the ages of a father and his son was 4 : 1. Eight years from now, the ratio of their ages will be 2 : 1. Find the present age of the father.
Verified Solution: Ratio 4 yrs ago = 4 : 1 (diff = 3). Ratio 8 yrs hence = 2 : 1 (diff = 1). Multiply second ratio by 3 to equalize age difference: 6 : 3 (diff = 3). Shift in units = 6 − 4 = 2 units = Total time gap (4 + 8 = 12 years) ⇒ 1 unit = 6 years. Father's age 4 yrs ago = 4 × 6 = 24 years ⇒ Present age = 24 + 4 = **28 years**.

Topic 10: Time and Work [Core Syllabus — High Priority]

Subtopics Covered & Rate Principle: Individual & combined work rates; **Why Work Rates Are Added (Not Days):** If A finishes a job in 20 days and B in 30 days, working together makes the job finish FASTER, not in 20 + 30 = 50 days! You must add the fraction of work done per day (1/20 + 1/30 = 5/60 = 1/12 of the job per day → 12 days) or take Total Work = LCM(20, 30) = 60 units and add daily efficiencies (3 + 2 = 5 units/day); Fractional work completion; Work with daily hours; Alternate-day working; Workers joining/leaving; Men-Women-Days ($M_1D_1H_1/W_1 = M_2D_2H_2/W_2$).

10.1. Fractional Work Rates, LCM Efficiency Method & Partial Work Completion

A. Concept (Class-10 Foundation): If a worker completes a fraction (m/n) of a work in t hours, then the time required to complete the WHOLE (100%) work alone is T = t × (n/m) hours. Once you have the full-work times T_A and T_B, define Total Work = LCM(T_A, T_B) units so all calculations become simple integers.

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Full-Work Time from Fraction:** If (m/n) work takes t hours, Full Work takes **T = t × (n/m)** hours. • **LCM Efficiency Method:** Total Work W = LCM(T_A, T_B); Efficiency E_A = W / T_A, E_B = W / T_B. • **Two-Worker Combined Time:** T_{A+B} = **$(T_A \times T_B) / (T_A + T_B)$** • **Time to Complete Fraction f of Work Together:** t_f = **$f \times W / (E_A + E_B)$**.	Problems stating 'C alone can do 2/10 of a work in 4 hours, D alone can do 2/5 in 5 hours, how long to complete 13/20 of the work together?'.

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB LDC Arithmetic Ability (Question 2)]:
C alone can do (2/10) of a work in 4 hours. D alone can do (2/5) of the same work in 5 hours. If they work together, then how many hours will they take to complete (13/20) of the same work?
(A) 5 hours **(B)** 7 hours **(C)** 8.5 hours **(D)** 6.5 hours

E. Step-by-Step Solution:

1. **Given Information:** C does 2/10 (= 1/5) work in 4 hours; D does 2/5 work in 5 hours; Target work fraction = 13/20.
2. **Required Quantity:** Hours taken by C and D together to complete 13/20 of the work.
3. **Formula / Method:** Scale up to full-work times for C and D, set Total Work = LCM = 100 units, compute hourly efficiencies, and divide target work (65 units) by combined efficiency.
4. **Substitution of Values:** C's full time = 4 × (5/1) = 20 hours. D's full time = 5 × (5/2) = 25/2 hours. Let Total Work = LCM(20, 25) = 100 units.
5. **Calculation:** C's efficiency = 100 ÷ 20 = 5 units/hr. D's efficiency = 100 ÷ (25/2) = 8 units/hr. Combined efficiency (C + D) = 5 + 8 = 13 units/hr. Target work = (13/20) × 100 = 65 units. Time required = 65 ÷ 13 = 5 hours.
6. **Final Answer: 5 hours (Option A)**

F. Shortcut Method (Speed Technique): Hourly rate of C = (1/5)/4 = 1/20; hourly rate of D = (2/5)/5 = 2/25. Together = 1/20 + 2/25 = 13/100 per hour. To do 13/20 (= 65/100) work takes 65/13 = 5 hours!

G. Common Trap & Distractor Alert: Adding 4 hours + 5 hours = 9 hours, or forgetting to multiply by the target fraction 13/20 at the end.

H. Additional Practice Variation & Verified Solution:
Question: [Verified PYQ — DSSSB Warder Q5] Working 6 hours a day, R1 can complete a piece of work in 18

days. Working 15 hours a day, R2 can complete the same work in 12 days. In how many days can both R1 and R2 together complete the same work by working 9 hours a day?
Verified Solution: Convert to total hours first: R1 takes 6 × 18 = 108 hours; R2 takes 15 × 12 = 180 hours. Let Total Work = LCM(108, 180) = 540 units. Efficiency of R1 = 540 ÷ 108 = 5 units/hr; Efficiency of R2 = 540 ÷ 180 = 3 units/hr. Together = 5 + 3 = 8 units/hr. Working 9 hours/day, they complete 8 × 9 = 72 units/day. Days required = 540 ÷ 72 = **7.5 days**.

10.2. Workers Joining/Leaving, Alternate Days & Men-Women-Days (MDH Formula)

A. Concept (Class-10 Foundation): When workers join or leave mid-job, track the units of work completed in each phase and subtract from Total Work. When two workers work on alternate days (A on Day 1, B on Day 2...), group 2 days into 1 cycle of work (E_A + E_B), take the largest integer number of cycles that does not exceed Total Work, and assign the remaining units to the worker whose turn comes next.

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Chain Rule (MDH / W): $(M_1 \times D_1 \times H_1) / W_1 = (M_2 \times D_2 \times H_2) / W_2$** • **Worker Leaving Early (t days before completion):** Add $(t \times E_{leaving})$ to Total Work and divide by combined efficiency of all workers. • **AND / OR Manpower Equivalence:** If 'm_1 men OR w_1 women finish in D days', then m_2 men AND w_2 women finish in **$D \div [(m_2/m_1) + (w_2/w_1)]$** days.	Questions mentioning 'A leaves 3 days before completion', 'work on alternate days starting with A', or '10 men can do a work in 12 days, 6 men can do it in how many days?'.

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB Warder Arithmetic Ability (Question 3)]:
10 men can do a piece of work in 12 days. 6 men can do the same work in how many days?
(A) 28 days **(B)** 20 days **(C)** 22 days **(D)** 24 days

E. Step-by-Step Solution:

1. **Given Information:** M_1 = 10 men, D_1 = 12 days; M_2 = 6 men.
2. **Required Quantity:** Number of days D_2 required by 6 men.
3. **Formula / Method:** Apply the inverse proportion manpower-days formula $M_1 \times D_1 = M_2 \times D_2$.
4. **Substitution of Values:** $10 \times 12 = 6 \times D_2 \Rightarrow 120$ man-days $= 6 \times D_2$.
5. **Calculation:** $D_2 = 120 \div 6 = 20$ days.
6. **Final Answer: 20 days (Option B)**

F. Shortcut Method (Speed Technique): Manpower drops to 6/10 = 3/5, so days multiply by 5/3: 12 × (5/3) = 20 days.

G. Common Trap & Distractor Alert: Using direct proportion (12 × 6 / 10 = 7.2 days) instead of inverse proportion.

H. Additional Practice Variation & Verified Solution:
Question: A and B can complete a task in 12 days and 18 days respectively. They begin working on alternate days starting with A. In how many days will the task be completed?
Verified Solution: Total Work = LCM(12, 18) = 36 units. E_A = 36/12 = 3 units/day; E_B = 36/18 = 2 units/day. In 1 cycle of 2 days (Day 1 A, Day 2 B), work done = 3 + 2 = 5 units. Multiply by 7 cycles (14 days): 7 × 5 = 35 units completed. Remaining work = 36 − 35 = 1 unit. On Day 15, it is A's turn (efficiency 3 units/day), who takes 1/3 day. Total time = **14⅓ days**.

Topic 11: Pipes and Cisterns [Core Syllabus — Medium Priority]

Subtopics Covered: Inlet pipes (positive filling rate +E) vs. Outlet pipes / Leaks (negative emptying rate −E); Two or three pipes operating simultaneously; Pipes opened or closed at staggered times; Partially filled tanks & finding tank capacity in litres from leak drainage rate.

11.1. Inlet & Outlet Pipes Operating Together and Staggered Pipe Opening/Closing

A. Concept (Class-10 Foundation): Pipes and Cisterns uses the exact same LCM method as Time and Work, with one crucial sign rule: an Inlet pipe that fills the tank has a POSITIVE efficiency (+E units/hr), whereas an Outlet pipe or leak that empties the tank has a NEGATIVE efficiency (−E units/hr). When pipes are opened at different hours, calculate the units filled in the first phase, subtract from Total Tank Capacity, and divide the remaining capacity by the new net efficiency.

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Tank Capacity (Total Work):** $W = LCM(T_1, T_2, T_3)$ • **Net Filling Efficiency (Inlets F_1, F_2 and Outlet F_3):** $E_{net} = E_1 + E_2 - E_3$ • **Two Inlets (A and B min):** $T = (A \times B) / (A + B)$ • **One Inlet (A hr) and One Leak/Outlet (B hr, where B > A):** $T_{fill} =$ **(A × B) / (B − A)**.	Questions stating 'Two pipes F1 and F2 fill in x and y hours, pipe F3 empties in z hours; F1 and F2 are opened first and after t hours F3 is also opened...'.

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB Junior Assistant Arithmetic Ability (Question 10)]: Two pipes F1 and F2 alone can fill an empty tank in 4.5 hours and 6 hours respectively. Pipe F3 can empty the completely filled tank in 12 hours. Firstly both pipes F1 and F2 are opened and after 1 hour pipe F3 is also opened. In how much total time will the tank be completely filled?
(A) 5 hours **(B)** 4 hours **(C)** 3 hours **(D)** 3.5 hours

E. Step-by-Step Solution:

1. **Given Information:** Inlet F1 = 4.5 hours (= 9/2 hrs); Inlet F2 = 6 hours; Outlet F3 = −12 hours. F1 and F2 run alone for the first 1 hour; then F1, F2, and F3 run together.
2. **Required Quantity:** Total time (in hours) from the start to fill the tank completely.
3. **Formula / Method:** Set Tank Capacity = LCM(9/2, 6, 12) = 36 units. Find hourly efficiencies, subtract the units filled in the first hour, and divide the remainder by the 3-pipe net rate.
4. **Substitution of Values:** $E_1 = 36 \div 4.5 = +8$ units/hr; $E_2 = 36 \div 6 = +6$ units/hr; $E_3 = -(36 \div 12) = -3$ units/hr.
5. **Calculation:** In the 1st hour (F1 + F2 only): Units filled = 8 + 6 = 14 units. Remaining capacity to fill = 36 − 14 = 22 units. When F3 is also opened, Net efficiency = 8 + 6 − 3 = 11 units/hr. Time to fill remaining 22 units = 22 ÷ 11 = 2 hours. Total time = 1 hour + 2 hours = 3 hours.
6. **Final Answer: 3 hours (Option C)**

F. Shortcut Method (Speed Technique): 36 units total → 14 units in hour 1 → 22 units left at 11 units/hr = 2 more hours → 1 + 2 = 3 hours!

G. Common Trap & Distractor Alert: Answering 2 hours (the remaining time after F3 was opened) instead of the total time 1 + 2 = 3 hours, or adding F3's efficiency (+3) instead of subtracting (−3).

H. Additional Practice Variation & Verified Solution:
Question: [Verified PYQ — DSSSB Warder Q15] Two pipes X and Y can individually fill a tank in 48 and 72 minutes respectively. If they are opened simultaneously, how long would it take for the tank to fill?

Verified Solution: Using Combined Time = (X × Y)/(X + Y) = (48 × 72)/(48 + 72) = 3456 / 120 = **28.8 minutes** (or LCM(48,72)=144 units; E_X=3, E_Y=2; 144/5 = 28.8 min).

Topic 12: Time, Speed and Distance [Core Syllabus — High Priority]

Subtopics Covered: Fundamental relation (Distance = Speed × Time); Unit conversion between km/h and m/s (×5/18 and ×18/5); Average speed for equal distances (Harmonic Mean) vs equal times (Arithmetic Mean); Inverse proportionality of speed and time when distance is constant; Relative speed in opposite (S_1 + S_2) and same ($|S_1 - S_2|$) directions; Meeting, overtaking, journeys with stops, and circular tracks.

12.1. Average Speed (Harmonic Mean for Equal Distances) & Constant-Distance Speed-Time Ratio

A. Concept (Class-10 Foundation): Average Speed is ALWAYS defined as Total Distance ÷ Total Time—it is NEVER the simple average of speeds unless the durations (times) are equal. When a person travels k equal distance stretches at speeds S_1, S_2, ..., S_k, assume each distance stretch equals LCM(S_1, S_2, ..., S_k) km so each time stretch t_i = D / S_i is a clean integer!

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Unit Conversion:** 1 km/h = **5/18 m/s**; 1 m/s = **18/5 km/h** • **General Average Speed:** S_{avg} = Total Distance ÷ Total Time • **2 Equal Distances at x and y:** S_{avg} = **2xy / (x + y)** • **3 Equal Distances at x, y, z:** S_{avg} = **3xyz / (xy + yz + zx)** • **Constant Distance Rule:** Speed ∝ 1/Time ⇒ $S_1 / S_2 = T_2 / T_1$ • **Journey with Stops:** Stoppage min per hour = **[($S_{without\ stops} - S_{with\ stops}$) / $S_{without\ stops}$] × 60 min**.	Problems stating 'travels four equal distances at speeds S_1, S_2, S_3, S_4' or 'runs at 50 km/h after repairing and 40 km/h before repairing over the same distance'.

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB LDC Arithmetic Ability (Question 4)]:
A man travels four equal distances of 40 km each. His speeds while travelling these stretches of distances were 10 km/hr, 20 km/hr, 30 km/hr and 40 km/hr respectively. What is his average speed for the whole journey?
(A) 19.2 km/hr **(B)** 21.4 km/hr **(C)** 17.8 km/hr **(D)** 22.8 km/hr

E. Step-by-Step Solution:

1. **Given Information:** 4 equal stretches of 40 km each (Total Distance = 4 × 40 = 160 km); Speeds = 10, 20, 30, and 40 km/hr.
2. **Required Quantity:** Average speed for the entire 160 km journey (in km/hr).
3. **Formula / Method:** Compute the time taken for each 40 km stretch (t_i = 40 / S_i), sum the four times, and divide Total Distance (160 km) by Total Time.
4. **Substitution of Values:** t_1 = 40/10 = 4 hrs; t_2 = 40/20 = 2 hrs; t_3 = 40/30 = 4/3 hrs; t_4 = 40/40 = 1 hr.
5. **Calculation:** Total Time = 4 + 2 + 1 + 4/3 = 7 + 4/3 = 25/3 hours. Average Speed = Total Distance ÷ Total Time = 160 ÷ (25/3) = (160 × 3) ÷ 25 = 480 ÷ 25 = 19.2 km/hr.
6. **Final Answer: 19.2 km/hr (Option A)**

F. Shortcut Method (Speed Technique): LCM Distance Trick: Ignore '40 km' and pick each stretch = LCM(10,20,30,40) = 120 km! Then Total Distance = 4 × 120 = 480 km, and times are 12 + 6 + 4 + 3 = 25 hrs → 480 / 25 = 19.2 km/hr with zero fractions!

G. Common Trap & Distractor Alert: Taking the simple arithmetic mean (10 + 20 + 30 + 40)/4 = 25 km/hr.

H. Additional Practice Variation & Verified Solution:
Question: (a) [Verified PYQ — DSSSB LDC Q7] A scooter runs at 50 km/h after repairing and 40 km/h before repairing. It covers a distance in 5 hours after repairing. How long will it take before repairing? (b) [Verified PYQ — DSSSB Warder Q4] A bus covers 12 km in 10 minutes. If its speed is decreased by 12 km/h, find the time taken to cover the same distance.
Verified Solution: (a) Speed ratio After : Before = 50 : 40 = 5 : 4 ⇒ Time ratio = 4 : 5. Since 4 units = 5 hours ⇒ 5 units = **25/4 hours**. (b) Initial speed = 12 km ÷ (10/60 hr) = 72 km/h. New speed = 72 − 12 = 60 km/h. At 60 km/h (1 km per minute), 12 km takes **12 minutes**.

12.2. Relative Speed (Opposite vs Same Direction), Meeting Time & Bridge Crossing

A. Concept (Class-10 Foundation): When two bodies move simultaneously toward each other (opposite directions), the distance between them closes at the sum of their speeds: S_rel = $S_1 + S_2$. When they move in the same direction, the gap changes at the difference of their speeds: S_rel = $|S_1 - S_2|$. Remember to convert km and minutes into meters and seconds when the final question asks for seconds to cross a bridge.

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Opposite Direction Relative Speed:** S_{rel} = $\mathbf{S_1 + S_2}$; Meeting Time t = Initial Gap / ($S_1 + S_2$) • **Same Direction Relative Speed (Chasing):** S_{rel} = $\mathbf{\lvert S_1 - S_2 \rvert}$; Catch-up Time t = Initial Gap / ($S_1 - S_2$) • **Man / Point Object Crossing a Bridge of Length L:** Time = L / S_{man}.	Problems where a car/train and a man start D km apart moving in opposite/same directions with a given speed ratio and meet after T minutes.

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB Junior Assistant Arithmetic Ability (Question 7)]:
A car and a man are travelling in opposite directions. The car is 150 km away from the man. The speeds of the car and the man are in the ratio 11 : 4 respectively. They meet after 100 minutes. How much time will that man take to cross a 600 meters long bridge?
(A) 120 seconds **(B)** 90 seconds **(C)** 135 seconds **(D)** 105 seconds

E. Step-by-Step Solution:

1. **Given Information:** Initial distance D = 150 km = 150,000 m; Meeting time t = 100 minutes = 6,000 seconds; Speed ratio S_car : S_man = 11 : 4 (opposite directions); Bridge length = 600 m.
2. **Required Quantity:** Time taken by the man to cross the 600 m bridge (in seconds).
3. **Formula / Method:** Find relative speed (S_car + S_man) in m/s from D/t, use the ratio 11 : 4 to extract S_man in m/s, and divide 600 m by S_man.
4. **Substitution of Values:** S_rel = 150,000 m ÷ 6,000 s = 25 m/s. Since S_car : S_man = 11 : 4, 15 ratio units = 25 m/s.
5. **Calculation:** Man's speed S_man = 4 units = (4 / 15) × 25 = 20 / 3 m/s. Time for man to cross 600 m bridge = Distance ÷ Speed = 600 ÷ (20 / 3) = (600 × 3) ÷ 20 = 30 × 3 = 90 seconds.
6. **Final Answer: 90 seconds (Option B)**

F. Shortcut Method (Speed Technique): Keep everything in m/s from Step 1: 150,000 m / 6,000 s = 25 m/s → Man's speed = (4/15)×25 = 20/3 m/s → 600 / (20/3) = 90 s!

G. Common Trap & Distractor Alert: Converting to km/h first and then making a conversion mistake back to m/s, or subtracting speeds (11x − 4x = 7x) when moving in opposite directions.

H. Additional Practice Variation & Verified Solution:
Question: [Verified PYQ — DSSSB Jr. Asst Q11] Two stations P and Q are 380 km apart. A train leaves P for Q and simultaneously another train leaves Q for P. Both trains meet after 19 hours. If the speed of the first train is 2 km/hr more than the second, find the speed of the slower train.
Verified Solution: Relative speed ($S_1 + S_2$) = 380 km ÷ 19 hrs = 20 km/hr. Given $S_1 - S_2 = 2$ km/hr. Slower train speed S_2 = (20 − 2) ÷ 2 = **9 km/hr**.

Topic 13: Trains [Core Syllabus — High Priority]

Subtopics Covered: Train crossing a stationary point object (pole, standing man, signal post → Distance = L_train); Train crossing a stationary extended object (platform, bridge, tunnel, stationary train → Distance = L_train + L_object); Two trains of lengths L_1 and L_2 crossing each other in opposite ($S_1 + S_2$) or same ($|S_1 - S_2|$) directions; Staggered departure train meetings & distance ratios.

13.1. Train Crossing Pole, Platform, or Another Moving Train

A. Concept (Class-10 Foundation): Unlike a car or a man (treated as a point of zero length), a train has a significant physical length L. Therefore, for a train to completely pass any object, the total distance covered is ALWAYS the sum of the lengths of the train and the object! When two trains of lengths L_1 and L_2 cross each other, Total Distance = $L_1 + L_2$ (regardless of whether they move in the same or opposite direction!), while Relative Speed is $S_1 + S_2$ (opposite) or $|S_1 - S_2|$ (same direction).

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Crossing a Pole / Standing Man:** Time t = L_{train} / S_{train} • **Crossing a Platform / Bridge of Length L_p:** Time t = $(L_{train} + L_p) / S_{train}$ • **Two Trains Crossing (Opposite Direction):** Time t = $(L_1 + L_2) / (S_1 + S_2)$ • **Two Trains Crossing (Same Direction):** Time t = $(L_1 + L_2) / \|S_1 - S_2\|$ • **Unit Check:** Always convert Lengths to meters (m), Time to seconds (s), and multiply m/s by **18/5** to get km/h.	Problems giving lengths of two trains in meters, crossing time in seconds, and asking for speed in km/h.

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB Warder Arithmetic Ability (Question 9)]:
A train of length 600 meters passes another train of length 1,000 meters moving in the opposite direction in 96 seconds. If the speed of both the trains is the same, then what is the speed of each train?
(A) 21 km/h **(B)** 30 km/h **(C)** 18 km/h **(D)** 24 km/h

E. Step-by-Step Solution:

1. **Given Information:** L_1 = 600 m, L_2 = 1,000 m; Time t = 96 seconds; Direction = Opposite; Speeds are equal ($S_1 = S_2 = S$).
2. **Required Quantity:** Speed S of each train in km/h.
3. **Formula / Method:** Compute relative speed (2S) in m/s as ($L_1 + L_2$) / t, convert to km/h by multiplying by 18/5, and divide by 2.
4. **Substitution of Values:** 2S (in m/s) = (600 + 1,000) ÷ 96 = 1,600 ÷ 96 = 50 / 3 m/s.
5. **Calculation:** Convert 2S to km/h: 2S = (50 / 3) × (18 / 5) = 10 × 6 = 60 km/h. Since both trains have equal speed, S = 60 ÷ 2 = 30 km/h.
6. **Final Answer: 30 km/h (Option B)**

F. Shortcut Method (Speed Technique): Total distance = 1600 m in 96 s → 100 m in 6 s → 50/3 m/s = 60 km/h combined → 30 km/h each!

G. Common Trap & Distractor Alert: Subtracting lengths (1000 − 600 = 400 m) or forgetting to divide the relative speed 60 km/h by 2.

H. Additional Practice Variation & Verified Solution:
Question: [Verified PYQ — DSSSB LDC Q15] From point E, at 6:30 pm, a train starts moving towards F at 60 km/hr. Another train starts from F at 7:30 pm towards E at 50 km/hr. Both trains meet at 11:30 pm at point G. What is the ratio of the distances EG and FG?
Verified Solution: Train 1 (from E) travels from 6:30 pm to 11:30 pm = 5 hours at 60 km/hr ⇒ Distance EG = 60 × 5 = 300 km. Train 2 (from F) travels from 7:30 pm to 11:30 pm = 4 hours at 50 km/hr ⇒ Distance FG = 50 × 4 = 200 km. Ratio EG : FG = 300 : 200 = **3 : 2**.

Topic 14: Boats and Streams [Core Syllabus — Medium Priority]

Subtopics Covered: Speed of boat in still water (u) and speed of water current/stream (v); Downstream speed (D = u + v) and Upstream speed (U = u − v); Finding u and v from D and U; Round-trip total time and equal-time distance equations.

14.1. Downstream & Upstream Speeds, Still-Water Conversion & Round-Trip Equations

A. Concept (Class-10 Foundation): When a boat rows WITH the water current (Downstream), the stream pushes the boat forward, so Downstream Speed D = u + v. When the boat rows AGAINST the current (Upstream), the stream resists the boat, so Upstream Speed U = u − v. Never confuse 'Downstream speed D' with 'Still-water speed u'!

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Downstream Speed (with current):** D = **u + v** • **Upstream Speed (against current):** U = **u − v** • **Still-Water Speed (from D and U):** u = **(D + U) / 2** • **Stream Speed (from D and U):** v = **(D − U) / 2** • **Round-Trip Time for Distance d:** Total Time T = **[d / (u + v)] + [d / (u − v)] = (2du) / (u^2 − v^2)**.	Questions mentioning 'speed of boat in still water', 'speed of stream/current', 'upstream', and 'downstream'.

D. Solved DSSSB-Style MCQ [DSSSB Section-A Standard Pattern Question (Boats & Streams)]:
A boat can row at a speed of 12 km/hr in still water, and the speed of the stream is 3 km/hr. How much total time will the boat take to go to a place 45 km downstream and return back to the starting point?
(A) 6 hours **(B)** 7.5 hours **(C)** 8 hours **(D)** 9 hours

E. Step-by-Step Solution:

1. **Given Information:** Speed in still water u = 12 km/hr; Stream speed v = 3 km/hr; One-way distance d = 45 km.
2. **Required Quantity:** Total round-trip time (Downstream time + Upstream time).
3. **Formula / Method:** Calculate D = u + v and U = u − v, then sum d/D and d/U.
4. **Substitution of Values:** Downstream speed D = 12 + 3 = 15 km/hr; Upstream speed U = 12 − 3 = 9 km/hr.
5. **Calculation:** Downstream time t_D = 45 ÷ 15 = 3 hours. Upstream time t_U = 45 ÷ 9 = 5 hours. Total round-trip time = 3 + 5 = 8 hours.
6. **Final Answer: 8 hours (Option C)**

F. Shortcut Method (Speed Technique): 45/15 + 45/9 = 3 + 5 = 8 hours.

G. Common Trap & Distractor Alert: Dividing total distance 90 km by still-water speed 12 km/hr (90 ÷ 12 = 7.5 hours, Option B)! Stream slows you down on the longer-duration upstream leg more than it helps on the shorter downstream leg.

H. Additional Practice Variation & Verified Solution:
Question: A swimmer rows 36 km downstream in 4 hours and 24 km upstream in 6 hours. Find the speed of the swimmer in still water and the speed of the current.
Verified Solution: Downstream speed D = 36 ÷ 4 = 9 km/hr. Upstream speed U = 24 ÷ 6 = 4 km/hr. Speed in still water u = (9 + 4)/2 = **6.5 km/hr**; Speed of current v = (9 − 4)/2 = **2.5 km/hr**.

Topic 15: Mixture and Alligation [Core Syllabus — Medium Priority]

Subtopics Covered: Adding an ingredient to shift the mixture ratio (unchanged component anchoring); Rule of Alligation (linking cheaper rate C, dearer rate D, and mean rate M); Mixing two vessels with different ratios; Repeated withdrawal and replacement of liquid.

15.1. Ratio Shift by Ingredient Addition, Alligation Cross-Rule & Repeated Replacement

A. Concept (Class-10 Foundation): When only ONE ingredient (say, sugar solution) is added to a mixture, the quantity of the OTHER ingredient (salt solution) remains strictly UNCHANGED! Equating the unchanged ingredient's ratio units before and after addition solves the problem in seconds. When two varieties at prices/concentrations C and D are mixed to produce a mean value M, the Rule of Alligation states that their quantities are in the ratio (D − M) : (M − C).

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Unchanged Component Principle:** If component A is added, Quantity of $B_{initial}$ = Quantity of B_{final}. • **Alligation Ratio:** (Quantity of Cheaper) / (Quantity of Dearer) = **(D − M) / (M − C)** • **Repeated Replacement (x litres drawn from V litres and replaced n times):** Final Pure Liquid Remaining = **$V \times [1 - (x / V)]^n$**.	Problems stating 'In a mixture of V litres, ratio is a : b; how much of the first liquid must be added to make the ratio c : d?' or '4 litres drawn and replaced with water twice'.

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB Warder Arithmetic Ability (Question 6)]:
In a mixture of 45 litres, the ratio of sugar solution to salt solution is 1 : 2. What is the amount of sugar solution to be added if the ratio has to be 2 : 1?
(A) 55 litres **(B)** 45 litres **(C)** 50 litres **(D)** 20 litres

E. Step-by-Step Solution:

1. **Given Information:** Total initial mixture = 45 litres; Initial ratio (Sugar : Salt) = 1 : 2; Target ratio (Sugar : Salt) = 2 : 1.
2. **Required Quantity:** Litres of sugar solution to be added.
3. **Formula / Method:** Split 45 L in the ratio 1 : 2 to find initial Sugar and Salt, keep Salt constant at 30 L, and find the new Sugar quantity required for a 2 : 1 ratio.
4. **Substitution of Values:** Initial Sugar = (1/3) × 45 = 15 litres; Initial Salt = (2/3) × 45 = 30 litres.
5. **Calculation:** Let x litres of sugar solution be added: (15 + x) / 30 = 2 / 1 ⇒ 15 + x = 60 ⇒ x = 60 − 15 = 45 litres.
6. **Final Answer: 45 litres (Option B)**

F. Shortcut Method (Speed Technique): Anchor Unchanged Salt: Initial Sugar : Salt = 1 : 2 (Total = 3 units = 45 L ⇒ 1 unit = 15 L). Target Sugar : Salt = 2 : 1 = 4 : 2 (making Salt = 2 units in both!). Sugar increases from 1 unit to 4 units = +3 units = 3 × 15 = 45 litres!

G. Common Trap & Distractor Alert: Thinking that flipping 1 : 2 to 2 : 1 requires adding only 15 litres (to go from 15 L to 30 L, which actually gives 30 : 30 = 1 : 1, not 2 : 1!).

H. Additional Practice Variation & Verified Solution:
Question: (a) In what ratio must rice at Rs. 42/kg be mixed with rice at Rs. 56/kg so that the mixture is worth Rs. 48/kg? (b) A container has 40 litres of pure milk. 4 litres of milk is taken out and replaced with water, and this process is repeated one more time. How much milk is left?
Verified Solution: (a) By Alligation: (Dearer − Mean) : (Mean − Cheaper) = (56 − 48) : (48 − 42) = 8 : 6 = **4 : 3**. (b) Final Milk = $40 \times (1 - 4/40)^2 = 40 \times (9/10)^2 = 40 \times (81/100)$ = **32.4 litres**.

Topic 16: Unitary Method and Work-Wages Problems [Core Syllabus — Medium Priority]

Subtopics Covered: Direct unitary proportion (Cost/Quantity/Boxes); Inverse unitary proportion (Workers & Days); Distribution of wages strictly in the ratio of total work contributed (or efficiency ratio when working together for the same duration).

16.1. Unitary Cost/Quantity Division & Wage Distribution by Work Contribution

A. Concept (Class-10 Foundation): The Unitary Method finds the value of 1 single unit by division, then scales up to k units by multiplication. In Work-and-Wages problems, money (wages) is ALWAYS divided in the ratio of the WORK DONE by each person ($W_1 : W_2 : W_3$)—never in the ratio of days taken! If all workers work for the exact same number of days, divide wages in the ratio of their daily efficiencies ($1/T_1 : 1/T_2 : 1/T_3$).

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Unitary Division:** Items per box = Total Items ÷ Number of Boxes • **Wages Rule (General):** $Wage_A : Wage_B$ = **(Work Done by A) : (Work Done by B)** • **Wages Rule (Working Together Full Time):** $Wage_A : Wage_B = E_A : E_B = (1/T_A) : (1/T_B) = T_B : T_A$.	Direct division/packing word problems or problems where A, B, and C undertake a job for Rs. X and ask for one person's wage share.

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB Warder Arithmetic Ability (Question 2)]:
8,064 apples were packed equally in 32 boxes. How many apples were packed in each box?
(A) 202 **(B)** 222 **(C)** 250 **(D)** 252

E. Step-by-Step Solution:

1. **Given Information:** Total apples = 8,064; Total boxes = 32.
2. **Required Quantity:** Number of apples in 1 box.
3. **Formula / Method:** Divide total apples by the number of boxes using factor cancellation (divide by 8, then by 4).
4. **Substitution of Values:** Apples per box = 8,064 ÷ 32.
5. **Calculation:** Divide numerator and denominator by 8: 8,064 ÷ 8 = 1,008 and 32 ÷ 8 = 4. Now divide 1,008 by 4: 1,008 ÷ 4 = 252 apples.
6. **Final Answer: 252 apples (Option D)**

F. Shortcut Method (Speed Technique): Estimate + Unit Digit: 32 × 250 = 8,000, and remaining 64 ÷ 32 = 2 ⇒ 250 + 2 = 252!

G. Common Trap & Distractor Alert: Making a long-division zero-dropping error (writing 8064 ÷ 8 = 108 instead of 1008).

H. Additional Practice Variation & Verified Solution:
Question: A alone can complete a work in 15 days and B alone in 20 days. With the help of C, they finish the work in 5 days and earn Rs. 3,600. Find C's share of the wages.
Verified Solution: Total Work = LCM(15, 20, 5) = 60 units. E_A = 60/15 = 4 units/day; E_B = 60/20 = 3 units/day; $E_{(A+B+C)}$ = 60/5 = 12 units/day. Thus C's efficiency E_C = 12 − (4 + 3) = 5 units/day. Since all three work for 5 days, wages split in efficiency ratio 4 : 3 : 5 (Total = 12 units). C's share = (5/12) × 3,600 = **Rs. 1,500**.

Topic 17: Mensuration: Two-Dimensional (2D) Shapes [Core Syllabus — High Priority]

Subtopics Covered: Square, Rectangle, Right/Isosceles/Scalene Triangle (Heron's formula & Pythagorean triplets), Equilateral Triangle, Regular Hexagon, Parallelogram, Rhombus, Trapezium, Circle, Semicircle, and Sector; Percentage change in 2D area when dimensions change; Circumradius and Inradius of triangles.

17.1. Triangles (Pythagorean Triplets, Heron's Formula, Circumradius) & Composite Quadrilaterals

A. Concept (Class-10 Foundation): Whenever a quadrilateral ABCD is given with all 4 side lengths and 1 diagonal AC, split ABCD into two triangles: ΔABC and ΔADC. ALWAYS check first if the three sides of either triangle form a Pythagorean Triplet ($a^2 + b^2 = c^2$), because the area of a right-angled triangle is simply ½ × base × perpendicular! For a general triangle with sides a, b, c, use Heron's formula.

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Right Triangle Area:** ½ × base × height; **Key Triplets:** (3,4,5), (5,12,13), (7,24,25), (8,15,17), **(9,40,41)**, (12,35,37), (20,21,29) • **Heron's Formula:** Area Δ = **√[s(s – a)(s – b)(s – c)]**, where semi-perimeter s = (a + b + c)/2 • **Equilateral Triangle (side a):** Area = **(√3 / 4)a^2**; Height h = (√3 / 2)a; Inradius r = a/(2√3); Circumradius R = a/√3 • **General Triangle Circumradius & Inradius:** R = **(abc) / (4Δ)**; r = **Δ / s**.	Quadrilateral with 4 sides and 1 diagonal given (e.g., 9, 40, 28, 15 and diagonal 41), or triangle with sides 12, 9, 9 asking for circumradius.

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB TGT Arithmetic Ability (Question 5)]:
ABCD is a quadrilateral in which AB = 9 cm, BC = 40 cm, CD = 28 cm, and AD = 15 cm. If diagonal AC = 41 cm, then the area (in cm^2) of the quadrilateral is:
(A) 390 **(B)** 270 **(C)** 282 **(D)** 306

E. Step-by-Step Solution:

1. **Given Information:** Quadrilateral ABCD split by diagonal AC = 41 cm into ΔABC (sides 9, 40, 41 cm) and ΔADC (sides 15, 28, 41 cm).
2. **Required Quantity:** Total Area of quadrilateral ABCD = Area(ΔABC) + Area(ΔADC) in cm^2.

3. **Formula / Method:** Recognize (9, 40, 41) as a right-angled Pythagorean triplet for ΔABC, and apply Heron's formula to ΔADC (sides 15, 28, 41).
4. **Substitution of Values:** In ΔABC: $9^2 + 40^2 = 81 + 1600 = 1681 = 41^2$, so ∠B = 90° and Area(ΔABC) = ½ × 9 × 40 = 180 cm². In ΔADC: s = (15 + 28 + 41)/2 = 84/2 = 42 cm.
5. **Calculation:** Area(ΔADC) = √[42 × (42 − 15) × (42 − 28) × (42 − 41)] = √[42 × 27 × 14 × 1] = √[(14 × 3) × (9 × 3) × 14] = 14 × 3 × 3 = 126 cm². Total Area = 180 + 126 = 306 cm².
6. **Final Answer: 306 cm² (Option D)**

F. Shortcut Method (Speed Technique): Memorizing the triplet (9, 40, 41) saves you from running Heron's formula on 9, 40, 41 (where s = 45)! Factoring under the square root (42×27×14 = $14^2 \times 9^2$) avoids multiplying huge numbers.

G. Common Trap & Distractor Alert: Trying to use ½ × d_1 × d_2 or failing to spot that ∠B = 90°.

H. Additional Practice Variation & Verified Solution:
Question: [Verified PYQ — DSSSB MTS Q5] The sides of a triangle are 12 cm, 9 cm and 9 cm. What is the radius of the circumcircle of this triangle?
Verified Solution: Semi-perimeter s = (12 + 9 + 9)/2 = 15 cm. Area Δ = √[15 × (15−12) × (15−9) × (15−9)] = √[15 × 3 × 6 × 6] = 6 × 3√5 = 18√5 cm². Circumradius R = (a × b × c) / (4 × Δ) = (12 × 9 × 9) / (4 × 18√5) = (3 × 9) / (2√5) = 27 / (2√5) = **(27√5) / 10 cm**.

17.2. Regular Hexagon, Parallelogram, Rhombus, Trapezium & Circle Percentage Area Change

A. Concept (Class-10 Foundation): A regular hexagon of side 'a' is composed of 6 congruent equilateral triangles of side 'a', so its area is 6 × (√3/4)a² = (3√3/2)a². When the radius (or side) of any 2D shape with a single length dimension (like a circle, square, or equilateral triangle) changes by r%, its Area changes by the 2-step successive percentage formula: [2r + (r²/100)]%.

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Regular Hexagon (side a):** Area = **6 × (√3 / 4)a^2 = (3√3 / 2)a^2**; Perimeter = 6a • **Parallelogram:** Area = **Base × Corresponding Altitude (b × h)** • **Rhombus:** Area = **½ × d_1 × d_2**; Side a = ½√(d_1^2 + d_2^2) • **Trapezium:** Area = **½ × (Sum of parallel sides) × height** • **Circle & Semicircle (radius r):** Circle Area = πr^2, Circumference = 2πr; Semicircle Perimeter = **(π + 2)r = (36/7)r**.	Questions asking for the area of a regular hexagon, percentage decrease in circle area when radius drops by r%, or base/altitude of a parallelogram.

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB PRT Arithmetic Ability (Question 15)]:
What is the area of a regular hexagon of side 6 cm?
(A) 108√3 cm² **(B)** 64√3 cm² **(C)** 54√3 cm² **(D)** 72√3 cm²

E. Step-by-Step Solution:

1. **Given Information:** Side of regular hexagon a = 6 cm.
2. **Required Quantity:** Area of the regular hexagon (in cm²).
3. **Formula / Method:** Multiply the area of one equilateral triangle ((√3/4)a²) by 6.
4. **Substitution of Values:** Area = 6 × (√3 / 4) × 6².
5. **Calculation:** Area = 6 × (√3 / 4) × 36 = 6 × 9√3 = 54√3 cm².
6. **Final Answer: 54√3 cm² (Option C)**

F. Shortcut Method (Speed Technique): Remember: for side a = 6, (3√3 / 2) × 36 = 54√3 cm² in 3 seconds!

G. Common Trap & Distractor Alert: Forgetting to divide by 4 (writing 6 × 36√3 or 3 × 36√3 = 108√3, Option A).

H. Additional Practice Variation & Verified Solution:
Question: (a) [Verified PYQ — DSSSB PRT Q7] The radius of a circle is decreased by 40%. By what percent will its area decrease? (b) [Verified PYQ — DSSSB MTS Q14] The area of a parallelogram is 72 cm² and its altitude is twice the corresponding base. What is the length of the base?
Verified Solution: (a) Using successive change for r² (−40% twice): Net % decrease = 40 + 40 − (40×40)/100 = 80 − 16 = **64%**. (b) Area = b × h = b × (2b) = 2b² = 72 ⇒ b² = 36 ⇒ b = **6 cm**.

Topic 18: Mensuration: Three-Dimensional (3D) Shapes [Core Syllabus — High Priority]

Subtopics Covered: Cube, Cuboid, Right Circular Cylinder, Right Circular Cone, Sphere, and Hemisphere; Volume, Curved/Lateral Surface Area (CSA/LSA), Total Surface Area (TSA), and Longest Diagonal; Cutting the largest 3D solid from another solid (e.g., cylinder/sphere from a cuboid/cube); Recasting/melting solids (Volume invariance) & multi-solid volume ratios.

18.1. Maximal Solid Cut Out of Cuboid/Cube & 3D Statement Verification (Cube / Hemisphere)

A. Concept (Class-10 Foundation): When a cylinder is cut out from a cuboid of dimensions l × b × h, there are up to 3 possible orientations (depending on which face acts as the circular base). For a circular base resting on a rectangular face of dimensions x × y, the maximum diameter of the circle cannot exceed the SMALLER side min(x, y), so max radius r = min(x, y)/2, and the cylinder's height is the third dimension z!

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Cube (edge a):** Volume = a^3; LSA = $4a^2$; TSA = $6a^2$; Diagonal = a√3 • **Cuboid (l, b, h):** Volume = lbh; LSA (4 walls) = 2(l+b)h; TSA = 2(lb+bh+hl); Diagonal = $\sqrt{(l^2+b^2+h^2)}$ • **Cylinder (r, h):** Volume = $\pi r^2 h$; CSA = 2πrh; TSA = 2πr(r + h) • **Hemisphere (radius r):** Volume = $\mathbf{(2/3)\pi r^3}$; CSA = $\mathbf{2\pi r^2}$; TSA = $\mathbf{3\pi r^2}$.	Questions asking for 'volume of the largest cylinder/sphere that can be cut from a cuboid/cube' or two-statement verification on a cube or hemisphere.

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB LDC Arithmetic Ability (Question 3)]:
The length, breadth and height of a solid cuboid are 3 cm, 4 cm and 5 cm respectively. What is the volume of the largest cylinder which can be cut from this cuboid?
(A) (264/7) cm³ **(B)** (1056/7) cm³ **(C)** (182/7) cm³ **(D)** (528/7) cm³

E. Step-by-Step Solution:

1. **Given Information:** Cuboid dimensions: l = 3 cm, b = 4 cm, h = 5 cm.
2. **Required Quantity:** Maximum possible volume of a right circular cylinder cut from the cuboid.
3. **Formula / Method:** Test the possible orientations by choosing which pair of dimensions forms the base (radius r = min(dim_1, dim_2)/2) and the remaining dimension as height H.
4. **Substitution of Values:** Case 1 (Base on 3×4 face, H = 5): r_1 = min(3,4)/2 = 1.5 cm, H_1 = 5 cm. Case 2 (Base on 4×5 face, H = 3): r_2 = min(4,5)/2 = 2 cm, H_2 = 3 cm. (Case 3 on 3×5 face gives r=1.5, H=4, clearly smaller than Case 1).

5. **Calculation:** Case 1 Volume V_1 = (22/7) × (3/2)² × 5 = 495/14 = 247.5/7 cm³. Case 2 Volume V_2 = (22/7) × 2² × 3 = (22 × 4 × 3)/7 = 264/7 cm³. Since 264/7 > 247.5/7, the largest cylinder has volume 264/7 cm³.
6. **Final Answer: (264/7) cm³ (Option A)**

F. Shortcut Method (Speed Technique): Because r is SQUARED in πr²H, maximizing radius r = 2 cm (from the 4×5 face) with H = 3 cm gives r²H = 4×3 = 12, beating r = 1.5 cm with H = 5 cm (r²H = 2.25×5 = 11.25)!

G. Common Trap & Distractor Alert: Assuming the longest side 5 cm must be the height of the cylinder (which gives 495/14 cm³ instead of 264/7 cm³), or taking radius = 4 cm instead of diameter = 4 cm (giving Option B = 1056/7).

H. Additional Practice Variation & Verified Solution:
Question: (a) [Verified PYQ — DSSSB LDC Q14] A hemisphere has radius 3.5 cm. Check Statements: (I) 6 × Volume = 542 cm³; (II) 4 × CSA = 154 cm². (b) [Verified PYQ — DSSSB Jr. Asst Q1] A cube has edge 8 cm. Check: (I) TSA is 2/3 of Volume; (II) 4 × Volume = 2096 cm³.
Verified Solution: (a) For r = 7/2 cm: 6 × Volume = 6 × (2/3) × (22/7) × (7/2)³ = 4 × 22 × (49/8) = **539 cm³** (≠ 542, Statement I False). 4 × CSA = 4 × 2 × (22/7) × (7/2)² = 8 × (77/2) = **308 cm²** (≠ 154, Statement II False). Thus **Neither I nor II**. (b) For a = 8 cm: TSA = 6×64 = 384, Volume = 512 ⇒ TSA/V = 384/512 = **3/4** (≠ 2/3, I False); 4×V = 4×512 = **2048 cm³** (≠ 2096, II False). Thus **Neither I nor II**.

18.2. Cone, Sphere & Cube Volume Ratios, Recasting Solids & Painting/Construction Cost

A. Concept (Class-10 Foundation): When comparing volumes of cones, spheres, cylinders, and cubes, write out the symbolic formulas first and cancel common factors before multiplying large numbers. When one 3D solid is melted and recast into N smaller solids, Total Volume Before = Total Volume After.

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Cone (radius r, height h, slant height l = √(r²+h²)):** Volume = **(1/3)πr²h**; CSA = **πrl**; TSA = **πr(r + l)** • **Sphere (radius r):** Volume = **(4/3)πr³**; Surface Area = **4πr²** • **Recasting / Melting Count:** Number of small objects N = $\text{Volume}_{\text{large}} \div \text{Volume}_{\text{small}}$.	Problems asking for the ratio of volumes of a cone, a sphere, and a cube, or the difference in volume between a sphere and a cone.

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB Junior Assistant Arithmetic Ability (Question 12)]:
A solid cone has radius and height as 3 cm and 7 cm respectively. A solid sphere has radius 10.5 cm. The edge of a cube is 6 cm. What is the ratio of the volumes of the cone, sphere and cube?
(A) 22 : 1217 : 36 **(B)** 33 : 1415 : 96 **(C)** 11 : 1123 : 64 **(D)** 22 : 1617 : 72

E. Step-by-Step Solution:

1. **Given Information:** Cone: r_1 = 3 cm, h_1 = 7 cm; Sphere: r_2 = 10.5 cm (= 21/2 cm); Cube: a = 6 cm.
2. **Required Quantity:** Ratio V_cone : V_sphere : V_cube in simplest integer form.
3. **Formula / Method:** Compute each volume using π = 22/7, then divide all three results by their HCF (3).
4. **Substitution of Values:** V_cone = (1/3) × (22/7) × 3² × 7 = 66 cm³. V_sphere = (4/3) × (22/7) × (21/2)³ = (4/3) × (22/7) × (9261/8) = 11 × 441 = 4,851 cm³. V_cube = 6³ = 216 cm³.
5. **Calculation:** Ratio = 66 : 4,851 : 216. Divide all three terms by 3: 66÷3 = 22; 4,851÷3 = 1,617; 216÷3 = 72. Thus the ratio is 22 : 1617 : 72.
6. **Final Answer: 22 : 1617 : 72 (Option D)**

F. Shortcut Method (Speed Technique): Compare just Cone : Cube first! V_cone = 66 and V_cube = 216 ⇒ 66 : 216 = 11 : 36 = 22 : 72. Look at the options: ONLY Option D has 22 as the first term and 72 as the third term! You don't even have to compute 10.5^3 for the sphere!

G. Common Trap & Distractor Alert: Wasting 90 seconds computing $10.5^3 = 1157.625$ by hand instead of checking the ratio of the two easy terms (Cone : Cube = 22 : 72) against the four options.

H. Additional Practice Variation & Verified Solution:
Question: [Verified PYQ — DSSSB JE Q8] The radius of a sphere is 7 cm. The radius of a cone is 7 cm and its height is 10 cm. Find the difference in their volumes (in cm^3).
Verified Solution: $V_{sphere} - V_{cone} = (4/3)\pi r^3 - (1/3)\pi r^2 h = (1/3)\pi r^2(4r - h)$. Substitute r = 7, h = 10: $(1/3) \times (22/7) \times 49 \times (4\times7 - 10) = (154 / 3) \times 18 = 154 \times 6 =$ **924 cm^3**.

Topic 19: Basic Algebra and Equations [Core Syllabus — Medium Priority]

Subtopics Covered: Linear equations in one and two variables (Heads & Legs / Two-type item word problems); Remainder and Factor Theorem for polynomials; Algebraic identities ($x + 1/x = k \Rightarrow x^2 + 1/x^2 = k^2 - 2$ and $x^3 + 1/x^3 = k^3 - 3k$).

19.1. Two-Variable Linear Word Problems (Heads & Legs) & Polynomial Factor Theorem

A. Concept (Class-10 Foundation): In classic two-type count problems (e.g., cows with 4 legs and hens with 2 legs, having H total heads and L total legs), if all H animals were 2-legged hens, there would be 2H legs. Every cow adds 2 extra legs! Thus, Number of 4-legged cows = (L − 2H) ÷ 2. In polynomial problems, the Factor Theorem states that if (x − a) is a factor of P(x), then substituting x = a gives P(a) = 0.

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Heads (H) & Legs (L) Shortcut (4-legged & 2-legged animals):** – **Number of 4-legged animals (Cows): (L − 2H) / 2 = (L/2) − H** – **Number of 2-legged birds (Hens): (4H − L) / 2 = 2H − (L/2)** • **Polynomial Factor & Remainder Theorem:** When P(x) is divided by (x − a), Remainder R = P(a). If (x − a) is an exact factor, **P(a) = 0** • **Reciprocal Power Identities:** If x + 1/x = k, then **$x^2 + 1/x^2 = k^2 - 2$** and **$x^3 + 1/x^3 = k^3 - 3k$**.	Problems giving total heads and total legs of cows/hens (or 4-wheelers/2-wheelers), or stating '(x − a) is a factor of polynomial P(x)'.

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB TGT Arithmetic Ability (Question 14)]:
Mukesh has some cows and some hens in his shed. The total number of legs is 92 and the total number of heads is 29. Find the total number of cows in the shed.
(A) 19 **(B)** 17 **(C)** 14 **(D)** 12

E. Step-by-Step Solution:

1. **Given Information:** Total legs L = 92; Total heads H = 29 (each cow has 4 legs and 1 head; each hen has 2 legs and 1 head).
2. **Required Quantity:** Number of cows c.
3. **Formula / Method:** Form linear equation 4c + 2(29 − c) = 92, or apply the Half-Legs minus Heads shortcut.
4. **Substitution of Values:** 4c + 58 − 2c = 92 ⇒ 2c = 92 − 58 = 34.

5. **Calculation:** c = 34 ÷ 2 = 17 cows (and 29 − 17 = 12 hens).
6. **Final Answer: 17 cows (Option B)**

F. Shortcut Method (Speed Technique): Cows = (Legs ÷ 2) − Heads = (92 ÷ 2) − 29 = 46 − 29 = 17 in 2 seconds!

G. Common Trap & Distractor Alert: Finding the number of hens (29 − 17 = 12, Option D) when the question asks for cows.

H. Additional Practice Variation & Verified Solution:
Question: [Verified PYQ — DSSSB PRT Q2] If (x − 3) is a factor of the polynomial ($x^2 + 4xp - 3p$), then what is the value of p?
Verified Solution: By the Factor Theorem, since (x − 3) is a factor, substitute x = 3 and equate to 0: $(3)^2 + 4(3)p - 3p = 0 \Rightarrow 9 + 12p - 3p = 0 \Rightarrow 9 + 9p = 0 \Rightarrow 9p = -9 \Rightarrow$ **p = −1**.

Topic 20: Data Interpretation (Tables, Bar Graphs, Pie Charts & Line Graphs) [Core Syllabus — High Priority]

Subtopics Covered & Pre-Calculation Reading Protocol: Tabular Data, Clustered Bar Graphs, Pie Charts (Percentage 100% and Degree 360°), and Line Graphs. **How to Read Charts Before Calculating:** (1) Check the **Title & Units** (e.g., 'in Rs. Lakhs' vs 'in Thousands'); (2) Check the **Axis Scale & Legend** (which bar is Investment vs Sales, or Income vs Expenditure); (3) Check any **Footnote Formula** (e.g., 'Savings = Income − Expenditure'); (4) Calculate only the specific rows/bars needed, using unit ratios or percentage differences BEFORE multiplying by the grand total!

20.1. Clustered Bar Graphs & Multi-Row Tables (Composite Expressions & Savings Comparisons)

A. Concept (Class-10 Foundation): In DSSSB CBT shifts, 2 to 3 questions out of 20 come from Data Interpretation. Often, a bar graph or table defines composite variables like J_1 (average of series 1), J_2 (average of series 2), and J_3 (difference in a specific year) and asks for $J_2 - J_1 + J_3$. Instead of dividing each sum by N separately, compute $(Sum_2 - Sum_1)/N + J_3$ to save time.

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Difference of Two Averages (over same N years):** $J_2 - J_1$ = **$(Sum_2 - Sum_1) / N$** = $(\Sigma(S_i - I_i)) / N$ • **Income-Expenditure-Savings Rule:** Savings = Income − Expenditure • **'X is what % less than Y': [(Y − X) / Y] × 100%** (Denominator is ALWAYS the quantity after 'than').	Bar graphs or tables over 4–5 years asking for a combination of averages and differences (like $J_2 - J_1 + J_3$) or percentage comparisons of savings.

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB TGT Arithmetic Ability (Question 4)]:
A bar graph shows the amounts invested (in lakhs) [2002: 120, 2003: 225, 2004: 375, 2005: 330, 2006: 525] and value of sales (in lakhs) [2002: 200, 2003: 300, 2004: 500, 2005: 400, 2006: 600] of a company over 5 years. If J1 = Average investment (2002–2006), J2 = Average sales (2002–2006), and J3 = Difference between investment and sales in 2005, what is the value of (J2 − J1 + J3)?
(A) 160 **(B)** 260 **(C)** 150 **(D)** 155

E. Step-by-Step Solution:

1. **Given Information:** Investment (2002–06): 120, 225, 375, 330, 525; Sales (2002–06): 200, 300, 500, 400, 600.

2. **Required Quantity:** Value of $J_2 - J_1 + J_3$.
3. **Formula / Method:** Compute J_1 = Sum(Investment)/5, J_2 = Sum(Sales)/5, and J_3 = Sales(2005) − Investment(2005).
4. **Substitution of Values:** Sum(Investment) = 120 + 225 + 375 + 330 + 525 = 1,575 ⇒ J_1 = 1,575 ÷ 5 = 315. Sum(Sales) = 200 + 300 + 500 + 400 + 600 = 2,000 ⇒ J_2 = 2,000 ÷ 5 = 400. In 2005, J_3 = 400 − 330 = 70.
5. **Calculation:** $J_2 - J_1 + J_3$ = 400 − 315 + 70 = 85 + 70 = 155.
6. **Final Answer: 155 (Option D)**

F. Shortcut Method (Speed Technique): Year-by-year gap (Sales − Investment): 80 + 75 + 125 + 70 + 75 = 425 ⇒ Average gap ($J_2 - J_1$) = 425 ÷ 5 = 85. Add J_3 (70): 85 + 70 = 155!

G. Common Trap & Distractor Alert: Misreading $J_2 - J_1$ as $J_1 - J_2$ or subtracting J_3 instead of adding it.

H. Additional Practice Variation & Verified Solution:
Question: [Verified PYQ — DSSSB PRT Q13] A bar graph shows monthly Income and Expenditure: Person A (Income 6000, Exp 4800), Person B (Income 7200, Exp 5400), Person C (Income 5600, Exp 3000). The difference between the savings of A and C is what percentage less than the savings of B?
Verified Solution: Savings: A = 6000−4800 = 1200; B = 7200−5400 = 1800; C = 5600−3000 = 2600. Difference between savings of A and C = 2600 − 1200 = 1400. Required % less than B (1800) = [(1800 − 1400)/1800] × 100% = (400/1800) × 100% = 200/9% = **22 2/9%**.

20.2. Pie Charts (Percentage & Central Angle Conversions) & Multi-Sector Ratios

A. Concept (Class-10 Foundation): A complete Pie Chart represents 100% of the data or 360° of central angle. Therefore, 1% = 3.6° and 5% = 18°. Never calculate the absolute value of each sector separately! Instead, perform addition, subtraction, or ratio operations directly on the sector percentages (or degrees) and multiply by the total only once at the very end.

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Percentage to Central Angle: 100% = 360° ⇒ 1% = 3.6° (or 5% = 18°)** • **Sector Difference Shortcut:** Value(S_1) − Value(S_2) = **[(% of S_1 − % of S_2) / 100] × Total** • **Sector Scaling Shortcut:** If (S_1 + S_3)% = Given Value V, then (S_2 + S_4) Value = **V × [(S_2 + S_4)% / (S_1 + S_3)%]**.	Pie charts showing percentage distribution across 5 to 7 categories (e.g., pulses P1–P5 or car manufacturing procedures J1–J7).

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB Junior Assistant Arithmetic Ability (Question 9)]:
A pie chart shows the percentage quantity sold of 5 types of pulses: P1 = 12%, P2 = 14%, P3 = 31%, P4 = 18%, and P5 = 25%. If the total quantity sold of P1 and P3 together is 1,247 kg, then how much quantity is sold of P2 and P4 together?
(A) 928 kg **(B)** 982 kg **(C)** 1,073 kg **(D)** 725 kg

E. Step-by-Step Solution:

1. **Given Information:** P1 = 12%, P2 = 14%, P3 = 31%, P4 = 18%, P5 = 25%; Quantity of (P1 + P3) = 1,247 kg.
2. **Required Quantity:** Combined quantity sold of (P2 + P4) in kg.
3. **Formula / Method:** Sum the percentages of (P1 + P3) = 43%, find the weight per 1%, and multiply by (P2 + P4)% = 32%.
4. **Substitution of Values:** (P1 + P3)% = 12% + 31% = 43% = 1,247 kg ⇒ 1% = 1,247 ÷ 43 = 29 kg.

5. **Calculation:** (P2 + P4)% = 14% + 18% = 32%. Quantity of (P2 + P4) = 32 × 29 = 928 kg.
6. **Final Answer: 928 kg (Option A)**

F. Shortcut Method (Speed Technique): Unit digit check: 1% = 1247 ÷ 43 = 29 (since 3 × 9 ends in 7). Then 32 × 29 must end in 2 × 9 = 8 → Option A (928 kg) is the ONLY option ending in 8!

G. Common Trap & Distractor Alert: Transposing digits to pick 982 kg (Option B) or computing the grand total 2,900 kg and individual sectors P2 and P4 separately.

H. Additional Practice Variation & Verified Solution:
Question: [Verified PYQ — DSSSB PRT Q14] A pie chart shows the time taken in 7 procedures (J1 to J7) to make a car as a percentage of total time (1,200 hours). If J1 = 14% and J4 = 8%, what is the difference between the time taken in J1 and J4?
Verified Solution: Percentage difference = J1 − J4 = 14% − 8% = 6%. Time difference = 6% of 1,200 hours = 6 × 12 = **72 hours**.

Topic 21: Additional Relevant Topics [Supplementary / Post-Dependent]

Syllabus Scope Note: The topics below — Arithmetic Progressions (AP), Elementary Statistics (Mean, Median, Mode), Basic Permutations/Combinations & Probability, and Elementary Trigonometric Identities — are primarily tested in **Section B of TGT Mathematics** or occasionally appear as 1 supplementary question in general Section-A shifts (as observed in DSSSB TGT Arithmetic Q15 and Adda247/Testbook TGT papers). Master Core Topics 1–20 first before revising these supplementary formulas.

21.1. Arithmetic Progressions (AP), Elementary Statistics, Probability & Trig Identities

A. Concept (Class-10 Foundation): In an Arithmetic Progression (AP) with first term 'a' and common difference 'd', the nth term is T_n = a + (n − 1)d and the sum of the first n terms is S_n = (n/2)[2a + (n − 1)d]. In Statistics, for any moderately skewed distribution, the empirical relation is Mode = 3 × Median − 2 × Mean. In Trigonometry, $\sin^2\alpha + \cos^2\alpha = 1$, $\sec^2\alpha - \tan^2\alpha = 1$, and $\text{cosec}^2\alpha - \cot^2\alpha = 1$.

B. Formula Box & Variable Definitions	C. Recognition Technique (Exam Trigger)
• **Arithmetic Progression (AP):** T_n = **a + (n − 1)d**; S_n = **(n/2)[2a + (n − 1)d] = (n/2)(a + l)** • **Statistics (Mean, Median, Mode):** Empirical Formula: **Mode = 3 × Median − 2 × Mean** • **Permutations & Combinations:** ${}^nP_r = n!/(n-r)!$; ${}^nC_r = n!/[r!(n-r)!]$ • **Probability:** P(E) = (Favorable Outcomes) / (Total Outcomes); $0 \le P(E) \le 1$; $P(E) + P(\text{not } E) = 1$ • **Pythagorean Trig Identities:** $\sin^2\alpha + \cos^2\alpha = 1$; $1 + \tan^2\alpha = \sec^2\alpha$; $1 + \cot^2\alpha = \text{cosec}^2\alpha$.	Questions asking for the sum of N terms of an AP, simplifying $(\sin\alpha + \text{cosec}\,\alpha)^2 + (\cos\alpha + \sec\alpha)^2$, or finding Mode from Mean and Median.

D. Solved DSSSB-Style MCQ [Verified PYQ — DSSSB TGT Maths / Numerical Series (Question 12)]:
What is the sum of the first 51 terms of an Arithmetic Progression (AP) whose 2nd term is 2 and 4th term is 8?
(A) 3,774 **(B)** 7,458 **(C)** 3,477 **(D)** 3,570

E. Step-by-Step Solution:

1. **Given Information:** Number of terms n = 51; 2nd term T_2 = a + d = 2; 4th term T_4 = a + 3d = 8.
2. **Required Quantity:** Sum of the first 51 terms (S_{51}).

3. **Formula / Method:** Find common difference d from $(T_4 - T_2)/2$, find first term a, and substitute into S_n = (n/2)[2a + (n − 1)d].
4. **Substitution of Values:** $2d = T_4 - T_2 = 8 - 2 = 6 \Rightarrow d = 3$. Then $a + 3 = 2 \Rightarrow a = -1$.
5. **Calculation:** $S_{51} = (51 / 2) \times [2(-1) + (51 - 1)(3)] = (51 / 2) \times [-2 + 150] = (51 / 2) \times 148 = 51 \times 74 = 3{,}774$.
6. **Final Answer: 3,774 (Option A)**

F. Shortcut Method (Speed Technique): Middle-Term Shortcut for Odd n: The average of the first 51 terms of an AP is simply its middle (26th) term! Since $T_2 = 2$ and $d = 3$, $T_{26} = T_2 + 24d = 2 + 24(3) = 74$. Then $S_{51} = 51 \times 74 = 3{,}774$!

G. Common Trap & Distractor Alert: Taking a = 2 (the 2nd term) as the first term instead of finding a = −1.

H. Additional Practice Variation & Verified Solution:
Question: (a) [Verified PYQ — DSSSB TGT Arithmetic Q15] Simplify: $(\sin \alpha + \operatorname{cosec} \alpha)^2 + (\cos \alpha + \sec \alpha)^2$. (b) [Adda247 DSSSB TGT PDF Q37] How many 3-digit numbers between 100 and 1000 can be formed with the digits 5, 6, 7, 8, 9 if repetition of digits is not allowed?
Verified Solution: (a) Expand both squares: $(\sin^2\alpha + \operatorname{cosec}^2\alpha + 2 \sin \alpha \operatorname{cosec} \alpha) + (\cos^2\alpha + \sec^2\alpha + 2 \cos \alpha \sec \alpha) = (\sin^2\alpha + \cos^2\alpha) + (1 + \cot^2\alpha) + 2(1) + (1 + \tan^2\alpha) + 2(1) = 1 + 1 + \cot^2\alpha + 2 + 1 + \tan^2\alpha + 2 =$ **$\tan^2\alpha + \cot^2\alpha + 7$**. (b) A number between 100 and 1000 has 3 digits. From 5 distinct digits {5,6,7,8,9} without repetition, the number of 3-digit arrangements is $5 \times 4 \times 3 =$ **60 numbers**.

Part IV: Compact Formula and Shortcut Reference Tables (Tables A–H)

Every value and shortcut below has been mathematically verified and includes explicit validity conditions so you never apply a special-case trick where it does not hold.

Table A: High-Frequency Fraction-to-Percentage & Decimal Reference Table

Fraction	Exact %	Decimal %	Derived Multiples (DSSSB Favorites)	Multiplier (Increase / Decrease)
1/2	50%	50.00%	3/2 = 150%	+50% → ×(3/2) | −50% → ×(1/2)
1/3	33⅓%	33.33%	2/3 = 66⅔% (66.67%)	+33⅓% → ×(4/3) | −33⅓% → ×(2/3)
1/4	25%	25.00%	3/4 = 75%	+25% → ×(5/4) | −25% → ×(3/4)
1/5	20%	20.00%	2/5 = 40%, 3/5 = 60%, 4/5 = 80%	+20% → ×(6/5) | −20% → ×(4/5)
1/6	16⅔%	16.67%	5/6 = 83⅓% (83.33%)	+16⅔% → ×(7/6) | −16⅔% → ×(5/6)
1/7	14 2/7%	14.28%	2/7 = 28 4/7%, 3/7 = 42 6/7%	+14 2/7% → ×(8/7) | −14 2/7% → ×(6/7)
1/8	12½%	12.50%	3/8 = 37.5%, 5/8 = 62.5%, 7/8 = 87.5%	+12.5% → ×(9/8) | −12.5% → ×(7/8)
1/9	11 1/9%	11.11%	2/9 = 22 2/9% (PRT Q13), 4/9 = 44.44%	+11 1/9% → ×(10/9) | −11 1/9% → ×(8/9)
1/10	10%	10.00%	3/10 = 30%, 7/10 = 70%, 9/10 = 90%	+10% → ×(11/10) | −10% → ×(9/10)
1/11	9 1/11%	9.09%	2/11 = 18.18%, 3/11 = 27.27%	+9 1/11% → ×(12/11) | −9 1/11% → ×(10/11)

Fraction	Exact %	Decimal %	Derived Multiples (DSSSB Favorites)	Multiplier (Increase / Decrease)
1/12	8⅓%	8.33%	5/12 = 41⅔%, 7/12 = 58⅓%, 11/12 = 91⅔%	+8⅓% → ×(13/12) | −8⅓% → ×(11/12) (TGT Q6)
1/15	6⅔%	6.67%	2/15 = 13⅓%, 4/15 = 26⅔%	+6⅔% → ×(16/15) | −6⅔% → ×(14/15)
1/16	6¼%	6.25%	3/16 = 18.75%, 9/16 = 56¼% (MTS Q12)	+56¼% → ×(25/16) | −6.25% → ×(15/16)
1/20	5%	5.00%	3/20 = 15% (LDC Q5), 19/20 = 95% (MTS Q10)	+15% → ×(23/20) | −5% → ×(19/20)
1/25	4%	4.00%	2/25 = 8%, 6/25 = 24%, 9/25 = 36%	+4% → ×(26/25) | −4% → ×(24/25)

Table B: Squares (1–30), Cubes (1–15) & Essential Irrational Roots

Squares 1 to 15	Squares 16 to 30	Cubes 1 to 15	Roots & Powers
$1^2 = 1$ | $2^2 = 4$ | $3^2 = 9$ $4^2 = 16$ | $5^2 = 25$ | $6^2 = 36$ $7^2 = 49$ | $8^2 = 64$ | $9^2 = 81$ $10^2 = 100$ | $11^2 = 121$ $12^2 = 144$ | $13^2 = 169$ $14^2 = 196$ | $15^2 = 225$	$16^2 = 256$ | $17^2 = 289$ $18^2 = 324$ | $19^2 = 361$ $20^2 = 400$ | $21^2 = 441$ $22^2 = 484$ | $23^2 = 529$ $24^2 = 576$ | $25^2 = 625$ $26^2 = 676$ | $27^2 = 729$ $28^2 = 784$ | $29^2 = 841$ | $30^2 = 900$	$1^3 = 1$ | $2^3 = 8$ | $3^3 = 27$ $4^3 = 64$ | $5^3 = 125$ | $6^3 = 216$ $7^3 = 343$ | $8^3 = 512$ | $9^3 = 729$ $10^3 = 1000$ | $11^3 = 1331$ $12^3 = 1728$ | $13^3 = 2197$ $14^3 = 2744$ | $15^3 = 3375$	$\sqrt{2} \approx 1.414$ | $\sqrt{3} \approx 1.732$ $\sqrt{5} \approx 2.236$ | $\sqrt{6} \approx 2.449$ $\sqrt{7} \approx 2.646$ | $\sqrt{10} \approx 3.162$ $2^5 = 32$, $2^6 = 64$, $2^{10} = 1024$ $3^4 = 81$, $3^5 = 243$, $3^6 = 729$ $5^4 = 625$, $7^4 = 2401$

Table C: Fast Multiplication, Division & Squaring Shortcuts (With Validity Conditions)

Operation	Exact Mental Rule	Worked Example & Limitation / Condition
Multiply by 5, 25, 125	×5 = ×(10/2); ×25 = ×(100/4); ×125 = ×(1000/8)	48 × 25 = 4800 / 4 = 1200; 72 × 125 = 72000 / 8 = 9000. *(Universal for all numbers)*
Multiply by 9, 99, 999	$N \times (10^k - 1) = (N \times 10^k) - N$	47 × 99 = 4700 − 47 = 4653. *(Universal for all real numbers)*
Multiply by 11	Keep outer digits, add adjacent pairs right-to-left (carry over if sum ≥ 10).	352 × 11 → [3][3+5][5+2][2] = 3872. *(Always carry tens digit when pair sum ≥ 10)*
Square Number Ending in 5	For $(10n + 5)^2$, write **n(n + 1)** followed by **25**.	85^2 → [8 × 9][25] = 7225; 11.5^2 → [11×12].25 = 132.25. *(Valid ONLY when last digit is 5)*
Product of Complementary Pairs	If tens digits are equal (n) and units digits sum to 10 (a+b=10): **[n(n+1)][a×b]** (2 slots for ab).	43 × 47 → [4×5][3×7] = 2021. *(Valid ONLY when tens match and unit digits sum to 10)*

Table D: High-Speed Percentage & Reverse-Base Shortcuts

Shortcut Type	Exact Formula / Rule	Application & Validity
Commutative Rule	x% of y = y% of x	32% of 25 = 25% of 32 = ¼ × 32 = 8 (universal).
2-Step Successive Change	[a + b + (ab / 100)]%	Valid for any 2 successive % changes (price, area, depreciation).
Symmetric +x% & −x%	Always a net **DECREASE of (x^2/100)%**	+20% then −20% → net −4% decrease.
Reverse Base Shift (A↔B)	+1/n increase ↔ −1/(n+1) decrease	+20% (1/5) ↔ −16⅔% (1/6); +25% (1/4) ↔ −20% (1/5); +50% (1/2) ↔ −33⅓% (1/3).

Table E: Speed Unit Conversion Reference (km/h ↔ m/s)

km/h to m/s (×5/18)	km/h to m/s (×5/18)	Fractional Speeds	Time & Distance Units
18 km/h = 5 m/s 36 km/h = 10 m/s 54 km/h = 15 m/s	72 km/h = 20 m/s 90 km/h = 25 m/s 108 km/h = 30 m/s	30 km/h = 25/3 m/s 45 km/h = 12.5 m/s 60 km/h = 50/3 m/s	1 km = 1000 m 1 hr = 60 min = 3600 s 1 m/s = 3.6 km/h

Table F: Work-Rate, Efficiency & Cistern Reference Table

Scenario	Formula / LCM Setup	Key Condition / Exam Rule
Individual Rate & Total Work	Total Work W = LCM(T_A, T_B); E_A = W / T_A	Never add days (T_A + T_B); always add efficiencies (E_A + E_B).
Two Workers / Pipes Together	T_{A+B} = (T_A × T_B) / (T_A + T_B)	For 1 Inlet (A) and 1 Outlet (B): T = (A × B) / (B − A).
Efficiency vs Time Ratio	If E_A : E_B = m : n, then T_A : T_B = n : m	Applies when both complete the exact same amount of work.
Manpower-Days-Hours (MDH)	(M_1 × D_1 × H_1) / W_1 = (M_2 × D_2 × H_2) / W_2	Work W_1, W_2 goes in the DENOMINATOR (length, items, consumption).

Table G: Master 2D & 3D Mensuration Formula Reference Table

Shape	Perimeter / Curved Surface Area (CSA)	Area / Total Surface Area (TSA) & Volume
Square (side a)	Perimeter = 4a	Area = a^2 = ½ d^2; Diagonal d = a√2
Rectangle (l, b)	Perimeter = 2(l + b)	Area = l × b; Diagonal d = √(l^2 + b^2)
Triangle (a, b, c)	Perimeter = a + b + c = 2s	Area = ½ bh = √[s(s−a)(s−b)(s−c)]; Equil. = (√3/4)a^2
Regular Hexagon (a)	Perimeter = 6a	Area = 6 × (√3/4)a^2 = (3√3 / 2)a^2
Parallelogram / Rhombus	Parallelogram = 2(a+b) | Rhombus = 4a	Parallelogram Area = b×h | Rhombus Area = ½ d_1d_2
Trapezium	Sum of all 4 sides	Area = ½ × (sum of parallel sides) × height
Circle / Semicircle (r)	Circle C = 2πr | Semicircle P = (π+2)r = (36/7)r	Circle Area = πr^2 | Semicircle Area = ½ πr^2

Shape	Perimeter / Curved Surface Area (CSA)	Area / Total Surface Area (TSA) & Volume
Cube (edge a)	LSA (4 walls) = $4a^2$	TSA = $6a^2$; Volume = a^3; Diagonal = $a\sqrt{3}$
Cuboid (l, b, h)	LSA (4 walls) = $2(l + b)h$	TSA = $2(lb + bh + hl)$; Volume = lbh; Diag. = $\sqrt{(l^2+b^2+h^2)}$
Cylinder (r, h)	CSA = $2\pi rh$	TSA = $2\pi r(r + h)$; Volume = $\pi r^2 h$
Cone (r, h, $l=\sqrt{(r^2+h^2)}$)	CSA = πrl	TSA = $\pi r(r + l)$; Volume = $(1/3)\pi r^2 h$
Sphere / Hemisphere (r)	Sphere SA = $4\pi r^2$ | Hemisphere CSA = $2\pi r^2$	Sphere V = $(4/3)\pi r^3$ | Hemi. TSA = $3\pi r^2$, V = $(2/3)\pi r^3$

Table H: DSSSB Arithmetic Error-Prevention Checklist

Error Category	Typical Mistake Made Under Pressure	Mandatory Verification Rule
1. Unit Conversion	Mixing km/h with meters and seconds in Train/Bridge problems; using 6 months as T=6 in SI/CI.	Convert speed to m/s (×5/18) before dividing meters; convert months to years (÷12) and minutes to hours (÷60).
2. Denominator / Base	Computing Profit % on SP instead of CP; dividing by wrong base in 'A is what % less than B'.	The number immediately after **'than'** or **'of'** is ALWAYS the denominator. Profit/Loss is on CP unless stated otherwise.
3. Vinculum & Signs	Treating $-\overline{16 - 48}$ as $-16 - 48 = -64$.	Rewrite any bar $\overline{a - b}$ immediately as a bracket $(a - b)$ before touching the sign outside.
4. Decimal Alignment	Taking HCF of 16.9 and 15.21 as HCF(169, 1521).	Pad trailing zeros (16.90 and 15.21) so all numbers have identical decimal places before removing the decimal point.
5. Target Variable Check	Stopping at x, CP (Rs. 1800 in TGT Q8), or remaining time instead of MP, total time, or digit sum.	Underline the LAST 4 words of the question stem ('sum of digits of x', 'marked price', 'slower train') before clicking.

Part V: Research-Based Question-Pattern Observations & Exam Strategy

- **Direct-Formula vs. Multi-Step Distribution:** Across our 75-question verified DSSSB sample, roughly **35% of questions** are 1-step direct applications (e.g., BODMAS, hexagon area, simple ratio, single bar-chart subtraction), while **65% are 2-to-3 step composite problems** (e.g., LCM with remainder 3 AND divisible by 67; MP→CP→SP chain; car-man relative speed followed by bridge crossing; 5-class overlapping averages).
- **Disguised Calculation Complexity:** DSSSB examiners frequently use intimidating numbers (such as $0.86^3 + 0.14^3$, $7^{29} - 7^{30} - 7^{31} + 7^{32}$, or 9, 40, 41 cm quadrilateral sides) that collapse to 5-second mental arithmetic once you spot the algebraic identity, common power factor, or Pythagorean triplet.
- **Section-Locking Strategy in DSSSB CBT:** In DSSSB online exams, sections are typically submitted sequentially (once you lock and submit the 20 Arithmetic questions to move to the next section, you cannot return). Therefore, use a **Two-Pass Strategy over 20 minutes**: Pass 1 (12 minutes) — solve the 12–14 direct and familiar questions (Simplification, DI, Ratio, Percentages, HCF/LCM); Pass 2 (8 minutes)

— tackle the 6–8 multi-step word problems (3D Mensuration, CI, Time & Work, Trains). Never spend more than 90 seconds on a single 1-mark question.

Part VI: Final Mixed Practice Test (25 Exam-Style MCQs)

Disclosure & Instructions: All 25 multiple-choice questions below are **newly created practice questions** modeled strictly on the verified DSSSB Section-A syllabus and observed question patterns across Easy (8 Qs), Moderate (12 Qs), and Challenging (5 Qs) levels. *Time Limit: 22 Minutes | Marking: +1.00 Correct, −0.25 Wrong.*

1. **[Number System — Divisibility]** If the 6-digit number 74865A is exactly divisible by 36, then what is the value of A?
   (A) 2 (B) 4 (C) 6 (D) 8
2. **[Number System — Factors]** What is the total number of positive divisors of 16 × 27 × 25?
   (A) 45 (B) 60 (C) 30 (D) 72
3. **[Simplification — VBODMAS]** Find the value of: $60 - [60 - (20 + 60) - 60 - (60 - \overline{20 - 60}) + 40]$.
   (A) 180 (B) 200 (C) 120 (D) 80
4. **[Simplification — Decimal Identity]** Evaluate: $[(0.73)^3 + (0.27)^3] \div [(7.3)^2 + (2.7)^2 - 73 \times 0.27]$.
   (A) 0.1 (B) 0.01 (C) 1 (D) 0.001
5. **[HCF and LCM — Decimals]** What is the Highest Common Factor (HCF) of 1.44, 1.80, and 1.92?
   (A) 0.12 (B) 1.2 (C) 0.06 (D) 0.24
6. **[HCF and LCM — Remainder Rule]** Find the least number which when divided by 12, 15, 20 and 30 leaves a remainder of 4 in each case, but is exactly divisible by 32.
   (A) 64 (B) 124 (C) 244 (D) 184
7. **[Ratio and Proportion — Total Balancing]** Rs. 45,000 is divided among A, B and C such that A's share is 1/2 of the total share of B and C, and B's share is 1/4 of the total share of A and C. What is the share of C?
   (A) Rs. 18,000 (B) Rs. 21,000 (C) Rs. 15,000 (D) Rs. 24,000
8. **[Ratio and Proportion — Coin Value]** A bag contains Re 1, Rs 2, and Rs 5 coins in the ratio 4 : 3 : 2. If the total value of the coins in the bag is Rs. 180, how many Rs 2 coins are there?
   (A) 18 (B) 27 (C) 36 (D) 24
9. **[Percentages — Price & Expenditure]** The price of sugar increases by 25%, while a family increases its expenditure on sugar by 15% only. By what percentage does its sugar consumption decrease?
   (A) 8% (B) 10% (C) 12% (D) 7.5%
10. **[Percentages — Successive Depreciation]** The value of a scooter depreciates at 10% per annum. If its present value is Rs. 50,000, what is the percentage decrease in its value after 2 years?
    (A) 20% (B) 19% (C) 18% (D) 21%
11. **[Profit, Loss & Discount — MP-CP-SP]** A trader buys goods at a 20% discount on the marked price and sells them at a 25% profit on his cost price. If his selling price is Rs. 1,500, what is the marked price?
    (A) Rs. 1,200 (B) Rs. 1,500 (C) Rs. 1,600 (D) Rs. 1,800
12. **[Profit & Loss — Articles per Rupee]** A vendor bought 15 lemons for a rupee. How many lemons must he sell for a rupee to earn a profit of 25%?
    (A) 10 (B) 12 (C) 11 (D) 9
13. **[Simple Interest — Step-Up Rate]** A borrower takes a loan on simple interest at 6% p.a. for the 1st year, with an increase of 0.5% in each subsequent year. If he pays Rs. 3,375 as interest after 4 years, find the

loan amount.
(A) Rs. 12,000 (B) Rs. 12,500 (C) Rs. 13,500 (D) Rs. 15,000

14. **[Compound Interest — CI vs SI Difference]** The simple interest and compound interest (compounded annually) on a certain sum for 2 years at the same rate are Rs. 800 and Rs. 848 respectively. Find the annual rate of interest.
(A) 10% (B) 12% (C) 8% (D) 15%
15. **[Average — Member Replacement]** The average marks of 20 students in a test increases by 2.5 when a student scoring 35 marks is replaced by a new student. What are the marks of the new student?
(A) 75 (B) 80 (C) 85 (D) 90
16. **[Problems on Ages]** Five years ago, the ratio of the ages of P and Q was 5 : 6. Three years from now, the ratio of their ages will be 7 : 8. What is the sum of their present ages?
(A) 44 years (B) 54 years (C) 50 years (D) 48 years
17. **[Time and Work — Fractional Rates]** A can do 1/4 of a work in 5 days and B can do 2/5 of the same work in 12 days. Working together, in how many days will they complete 2/3 of the work?
(A) 8 days (B) 9 days (C) 10 days (D) 7.5 days
18. **[Pipes and Cisterns]** Pipes A and B can fill a tank in 6 hours and 9 hours respectively, while Pipe C can empty the full tank in 18 hours. If A and B are opened for 2 hours and then C is also opened, what is the total time taken to fill the tank?
(A) 3 hours (B) 3.5 hours (C) 4 hours (D) 4.5 hours
19. **[Time, Speed & Distance — Harmonic Mean]** A car covers three equal stretches of 60 km each at speeds of 20 km/hr, 30 km/hr, and 60 km/hr respectively. What is its average speed for the whole journey?
(A) 30 km/hr (B) 36 km/hr (C) 33.33 km/hr (D) 40 km/hr
20. **[Trains — Opposite Crossing]** A 400 m long train passes another 500 m long train moving in the opposite direction in 36 seconds. If the first train runs at 42 km/hr, what is the speed of the second train?
(A) 48 km/hr (B) 54 km/hr (C) 36 km/hr (D) 45 km/hr
21. **[Mixture & Alligation]** In a 60-litre mixture, the ratio of milk to water is 2 : 1. How many litres of water must be added to make the ratio of milk to water 1 : 2?
(A) 40 litres (B) 60 litres (C) 50 litres (D) 30 litres
22. **[Mensuration 2D — Right Triangle]** The sides of a triangle are 8 cm, 15 cm, and 17 cm. What is its area (in cm^2)?
(A) 60 cm^2 (B) 68 cm^2 (C) 120 cm^2 (D) 75 cm^2
23. **[Mensuration 3D — Cylinder from Cube]** What is the volume (in cm^3) of the largest right circular cylinder that can be cut from a solid cube of edge 14 cm? (Use $\pi = 22/7$)
(A) 2,156 cm^3 (B) 1,078 cm^3 (C) 2,744 cm^3 (D) 1,540 cm^3
24. **[Basic Algebra — Heads & Wheels]** A parking lot has auto-rickshaws (3 wheels) and cars (4 wheels). If there are 40 vehicles in total and 145 wheels in all, how many cars are in the lot?
(A) 15 (B) 25 (C) 20 (D) 18
25. **[Data Interpretation — Pie Chart]** In a pie chart of 2,500 students across 4 houses (Red: 28%, Blue: 22%, Green: 32%, Yellow: 18%), by how many students does Green house exceed Yellow house?
(A) 300 (B) 350 (C) 400 (D) 250

Part VII: Separate Answer Key & Complete Step-by-Step Worked Solutions

Q1–Q5	Q6–Q10	Q11–Q15	Q16–Q20	Q21–Q25
1. (C) 6	**6.** (A) 64	**11.** (B) Rs. 1,500	**16.** (B) 54 yrs	**21.** (B) 60 L
2. (B) 60	**7.** (B) Rs. 21,000	**12.** (B) 12	**17.** (A) 8 days	**22.** (A) 60 cm^2
3. (B) 200	**8.** (B) 27	**13.** (B) Rs. 12,500	**18.** (C) 4 hours	**23.** (A) 2,156 cm^3
4. (B) 0.01	**9.** (A) 8%	**14.** (B) 12%	**19.** (A) 30 km/hr	**24.** (B) 25
5. (A) 0.12	**10.** (B) 19%	**15.** (C) 85	**20.** (A) 48 km/hr	**25.** (B) 350

Step-by-Step Worked Solutions (With Topic & Method Used)

1. **[Topic 1: Number System — Co-Prime Divisibility Rule]** 36 = 4 × 9. Sum of digits of 74865A = 7 + 4 + 8 + 6 + 5 + A = 30 + A. For divisibility by 9, 30 + A = 36 ⇒ A = 6. Check divisibility by 4: last two digits = 56, and 56 ÷ 4 = 14 (exact). Answer: **6 (Option C)**.
2. **[Topic 1: Number System — Prime Factorisation Divisor Formula]** Write in prime bases: 16 = 2^4, 27 = 3^3, 25 = 5^2. Total positive divisors = (4 + 1)(3 + 1)(2 + 1) = 5 × 4 × 3 = **60 (Option B)**.
3. **[Topic 2: Simplification — VBODMAS & Vinculum]** Evaluate bar first: $\overline{20 - 60}$ = −40 ⇒ (60 − (−40)) = 100. Also (20 + 60) = 80. Inside square bracket: [60 − 80 − 60 − 100 + 40] = −140. Outer: 60 − (−140) = **200 (Option B)**.
4. **[Topic 2: Simplification — Cubic Identity]** Let a = 0.73, b = 0.27 (so a + b = 1.00). Denominator has 7.3 = 10a and 2.7 = 10b, scaling it by 10^2 = 100. Value = (a + b) / 100 = 1 / 100 = **0.01 (Option B)**.
5. **[Topic 3: HCF and LCM — Decimal HCF]** Multiply by 100: integers are 144, 180, 192. Smallest difference = 192 − 180 = 12, and 12 divides 144 (12×12), 180 (12×15), and 192 (12×16). Divide by 100: HCF = **0.12 (Option A)**.
6. **[Topic 3: HCF and LCM — Remainder + Exact Divisor]** LCM(12, 15, 20, 30) = 60. General form = 60k + 4. For k = 1, 60(1) + 4 = 64, which is divisible by 32 (64 ÷ 32 = 2). Answer: **64 (Option A)**.
7. **[Topic 4: Ratio & Proportion — Total Sum Balancing]** A : (B+C) = 1 : 2 ⇒ A = 1/3 of total = Rs. 15,000. B : (A+C) = 1 : 4 ⇒ B = 1/5 of total = Rs. 9,000. Thus C = 45,000 − (15,000 + 9,000) = **Rs. 21,000 (Option B)**.
8. **[Topic 4: Ratio & Proportion — Coin Denomination Value]** Let counts be 4x, 3x, 2x. Total value = 4x(1) + 3x(2) + 2x(5) = 20x = 180 ⇒ x = 9. Number of Rs 2 coins = 3x = 3 × 9 = **27 (Option B)**.
9. **[Topic 5: Percentages — Expenditure-Price-Consumption]** Start at 100: New Price = 125, New Expenditure = 115. Consumption drop = (125 − 115) / 125 × 100% = (10 / 125) × 100% = **8% (Option A)**.
10. **[Topic 5: Percentages — 2-Year Depreciation]** Using successive change for −10% twice: Net % decrease = 10 + 10 − (10 × 10)/100 = 20 − 1 = **19% (Option B)**.
11. **[Topic 6: Profit, Loss & Discount — MP-CP-SP Chain]** Trader's CP = 0.80 × MP. Trader's SP = 1.25 × CP = 1.25 × 0.80 × MP = 1.00 × MP = Rs. 1,500 ⇒ MP = **Rs. 1,500 (Option B)**.
12. **[Topic 6: Profit & Loss — Articles per Rupee]** Price factor increases from 100% to 125% (ratio 4 : 5), so quantity per rupee changes in inverse ratio 5 : 4. Since 5 units = 15 lemons ⇒ 4 units = **12 lemons (Option B)**.
13. **[Topic 7: Simple Interest — Step-Up Annual Rate]** Cumulative 4-year SI rate = 6% + 6.5% + 7% + 7.5% = 27%. Given 27% of P = 3,375 ⇒ P = (3,375 × 100) / 27 = 125 × 100 = **Rs. 12,500 (Option B)**.
14. **[Topic 7: Compound Interest — 2-Year CI vs SI Difference]** 1-year SI = 800 ÷ 2 = Rs. 400. Extra 2nd-year CI = 848 − 800 = Rs. 48. Annual rate R = (48 / 400) × 100% = **12% (Option B)**.
15. **[Topic 8: Average — Member Replacement]** New Student's Marks = Old Marks + (N × ΔAverage) = 35 + (20 × 2.5) = 35 + 50 = **85 (Option C)**.
16. **[Topic 9: Problems on Ages — Constant Difference Method]** Ratio 5 yrs ago = 5 : 6 (diff = 1); Ratio 3 yrs hence = 7 : 8 (diff = 1). Shift = 7 − 5 = 2 units = 8 years ⇒ 1 unit = 4 years. Sum 5 yrs ago = 11 × 4 = 44

years ⇒ Present sum = 44 + 2(5) = **54 years (Option B)**.

17. **[Topic 10: Time and Work — Fractional Rates & LCM]** Full-work times: A = 5 × 4 = 20 days; B = 12 × (5/2) = 30 days. Let Total Work = LCM(20, 30) = 60 units ⇒ E_A = 3, E_B = 2 (combined = 5 units/day). Time for 2/3 of 60 (= 40 units) = 40 ÷ 5 = **8 days (Option A)**.
18. **[Topic 11: Pipes and Cisterns — Staggered Opening]** Tank Capacity = LCM(6, 9, 18) = 18 units ⇒ E_A = +3, E_B = +2, E_C = −1. In first 2 hours (A+B), units filled = 2 × (3 + 2) = 10 units. Remaining 8 units filled at net rate (3 + 2 − 1 = 4 units/hr) takes 8 ÷ 4 = 2 hours. Total time = 2 + 2 = **4 hours (Option C)**.
19. **[Topic 12: Time, Speed & Distance — Harmonic Mean]** Total distance = 3 × 60 = 180 km. Times: t_1 = 60/20 = 3 hrs, t_2 = 60/30 = 2 hrs, t_3 = 60/60 = 1 hr (Total time = 6 hrs). Average speed = 180 ÷ 6 = **30 km/hr (Option A)**.
20. **[Topic 13: Trains — Opposite Crossing]** Relative speed = (400 + 500) m ÷ 36 s = 900 / 36 = 25 m/s = 25 × (18/5) = 90 km/hr. Since $S_1 + S_2 = 90$ and $S_1 = 42$ km/hr ⇒ $S_2 = 90 - 42$ = **48 km/hr (Option A)**.
21. **[Topic 15: Mixture & Alligation — Unchanged Component]** Initially Milk = 40 L, Water = 20 L. Milk stays unchanged at 40 L. For final ratio Milk : Water = 1 : 2, final Water must be 2 × 40 = 80 L. Water added = 80 − 20 = **60 litres (Option B)**.
22. **[Topic 17: Mensuration 2D — Pythagorean Triplet]** Since $8^2 + 15^2 = 64 + 225 = 289 = 17^2$, (8, 15, 17) is a right-angled triangle! Area = ½ × 8 × 15 = **60 cm² (Option A)**.
23. **[Topic 18: Mensuration 3D — Cylinder from Cube]** Max cylinder inside a cube of edge 14 cm has diameter = 14 cm (radius r = 7 cm) and height h = 14 cm. Volume = $\pi r^2 h$ = (22/7) × 7^2 × 14 = 22 × 7 × 14 = **2,156 cm³ (Option A)**.
24. **[Topic 19: Basic Algebra — All-3-Wheelers Shortcut]** If all 40 vehicles were 3-wheelers, there would be 3 × 40 = 120 wheels. Each car adds 1 extra wheel: Number of cars = 145 − 120 = **25 cars (Option B)**.
25. **[Topic 20: Data Interpretation — Pie Chart Difference]** Percentage difference (Green − Yellow) = 32% − 18% = 14%. Student difference = 14% of 2,500 = 14 × 25 = **350 students (Option B)**.

Part VIII: One-Page Rapid-Revision Formula Sheet, Exam-Day Checklist & Sources

Rapid-Revision Formula Summary (Topics 1–10)	Rapid-Revision Formula Summary (Topics 11–21)
• **Divisibility:** 36 = 4×9; 72 = 8×9; 88 = 8×11; 99 = 9×11. • **Factors of N = $2^p a^q b^r$:** Total = (p+1)(q+1)(r+1); Odd = (q+1)(r+1); Even = p(q+1)(r+1). • **Cubic Identities:** $(a^3+b^3)/(a^2-ab+b^2) = a+b$; $(a^3+b^3+c^3-3abc)/(a^2+b^2+c^2-ab-bc-ca) = a+b+c$. • **HCF & LCM:** $N_1 \times N_2$ = HCF×LCM; Least no. leaving rem r = LCM·k + r; Same unknown rem divisor = HCF($\lvert b-a \rvert$, $\lvert c-b \rvert$). • **Percentages:** Net change = a + b + ab/100%; Price +r% with fixed exp → Cons. drop = [r/(100+r)]×100%. • **Profit/Discount:** MP/CP = (100 + Profit%)/(100 − Discount%); False weight W g/kg → Profit% = [(1000−W)/W]×100%. • **Interest:** 2-yr (CI − SI) = $P(R/100)^2$ = (R/200)×SI_2; Half-yearly: r = R/2, n = 2T. • **Average & Ages:** Replacement New = Old ± N·ΔA; Age diff (b−a) is constant across all years. • **Time & Work:** W = LCM(T_1, T_2); $(M_1D_1H_1)/W_1 = (M_2D_2H_2)/W_2$.	• **Pipes:** Net Rate = $E_{in1} + E_{in2} - E_{out}$. • **Speed & Distance:** km/h × (5/18) = m/s; Equal dist. $S_{avg} = 2xy/(x+y)$; Constant D ⇒ $S_1/S_2 = T_2/T_1$. • **Trains:** Crossing time = ($L_1 + L_2$) / S_{rel}, where $S_{rel} = S_1 + S_2$ (opposite) or $\lvert S_1 - S_2 \rvert$ (same). • **Boats:** D = u + v, U = u − v ⇒ u = (D+U)/2, v = (D−U)/2. • **Mixture:** Alligation (D−M):(M−C); Replacement = $V(1 - x/V)^n$. • **2D Mensuration:** Triplets (3,4,5), (5,12,13), (8,15,17), (9,40,41); Equil. Δ = (√3/4)a^2; Hexagon = 6×(√3/4)a^2; Circle r −x% → Area −(2x − x^2/100)%. • **3D Mensuration:** Cube TSA=6a^2, V=a^3; Cylinder V=$\pi r^2 h$; Cone V=(1/3)$\pi r^2 h$; Sphere V=(4/3)πr^3; Hemi. CSA=2πr^2, TSA=3πr^2, V=(2/3)πr^3. • **Algebra & DI:** 4-legged count = (Legs/2) − Heads; Pie Chart 1% = 3.6°; AP S_n = (n/2)[2a + (n−1)d].

Final Exam-Day Action Checklist for DSSSB Section-A Arithmetic

1. **Budget Exactly 20 Minutes:** Do not let a single stubborn calculation steal time from your 100-mark Section-B post-specific paper.
2. **Scan Options Before Multiplying:** Use unit digits, digital roots (divisibility by 9 or 3), and extreme-term ratios (like Cone : Cube = 22 : 72 in Jr. Asst Q12) to eliminate 3 options before doing full multiplication.
3. **Guard Against Negative Marking (−0.25):** Never blind-guess when all 4 options look plausible; only make an educated guess if you have mathematically eliminated at least 2 options.
4. **Re-verify the Final Question Word:** Before clicking 'Save & Next', check whether the question asked for *Cost Price* vs *Marked Price*, *Cows* vs *Hens*, *Remaining Time* vs *Total Time*, or *Sum of Digits of x*.
5. **Standardize Rough-Sheet Layout:** Divide your CBT rough pad into 20 numbered boxes for the 20 Arithmetic questions so that if you skip a multi-step question on Pass 1, your LCM or ratio setup is already cleanly written when you return on Pass 2.
6. **Use Dimensional & Parity Sanity Checks:** Before locking an answer in Mensuration or Time-Speed-Distance, verify whether the answer must be a multiple of 11 (from π = 22/7), a multiple of 3 or 9, or strictly less than the harmonic bound (for average speed).

Verified Official & Secondary Source Links

- **Official DSSSB Portal (Notifications, Schemes & Answer Keys):** https://dsssb.delhi.gov.in/
- **DSSSB TGT Syllabus & Exam Pattern (2026 Reference):** https://testbook.com/dsssb-tgt/syllabus-exam-pattern and Drishti Teaching Exams DSSSB TGT Guide
- **DSSSB Official One-Tier Syllabus Wording:** https://byjus.com/govt-exams/dsssb-ae-syllabus/
- **Verified DSSSB Previous-Year Solved Question Banks Examined:** DSSSB TGT Arithmetic PYQs, DSSSB PRT Arithmetic PYQs, DSSSB LDC Arithmetic PYQs, DSSSB Junior Assistant Arithmetic PYQs, DSSSB MTS Arithmetic PYQs, and DSSSB Warder Arithmetic PYQs.